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Co-design approaches involving older adults in the development of electronic healthcare tools: a systematic review

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Co-design approaches involving older adults in the development of electronic healthcare tools: a systematic review

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Abstract

Objective

The objective of this systematic review is to synthesize the evidence regarding the current state of utilizing co-design approaches involving older adults when developing electronic healthcare tools (EHTs); how these approaches are defined, and at what stages they are employed. Our aim is to evaluate the extent to which co-design approaches engage older adults in the various involvement processes and levels of participation in the development lifecycle of EHTs.

Methods

PubMed, Embase, and Scopus databases were searched for studies from January 2010 to March 2021. Inclusion criteria were studies that employed co-design approaches in the development of EHTs, and the study population was aged 60 years and older.

Results

Twenty-five studies met the inclusion criteria for this review. Each study indicated that EHTs were designed with involvement from older adults. Heterogeneity with the terminology, frameworks, involvement processes, and levels of participation was discovered and was challenging to categorize. However, evidence was extracted from each study and mapped against two frameworks, (1) involvement processes and (2) levels of participation. EHTs were classified according to their associated health sector and defined by the healthcare needs in which they address. Lastly, subjective, and objective assessments were categorized by measure, endpoint, and validity.

Conclusions

Evidence suggests that co-design approaches aim to involve end-users in the development of EHTs. However, there are gaps in the evidence to support the levels at which older adults should be involved and the extent to which this involvement can increase the user's knowledge of the technology, either limiting or encouraging their collaboration and empowerment. Additionally, gaps exist around which measures should be employed when conducting user

testing with older adults. To ensure studies are optimizing the involvement of older adults when developing EHTs, improving existing co-design frameworks to meet the needs of older adults should be evaluated.

Systematic Review Registration

PROSPERO CRD42021240013

Strengths and limitations of this study

- This review provides a comprehensive assessment of the utilization of co-design approaches in studies that engage older adults in developing electronic healthcare tools.
- Undertook a review in multiple databases and developed a broad search strategy to ensure all "collaborative design" approaches were captured in this review.
- Utilized widely reported classification methods to categorize the included studies.
- Included only studies that indicated that older adults were involved in the development of an electronic healthcare tool (EHT).
- Findings are limited to the levels of participation, and involvement processes; further cross-classification of these variables was not conducted, as we were unable to specify the point within the study that a particular involvement process was employed.

Introduction

Co-design aims to elicit ideas and foster a non-hierarchical environment in which concepts, tacit knowledge, and lived experiences of all stakeholders, including the target population can be applied to develop tools that meet the needs of this population(1,2). Co-design is also likened to participatory design, each defined as focusing on the engagement and creativity of stakeholders, or those untrained in design efforts, but through collaborative efforts bring the representative interest into the development process (3,4). The basic principles that encapsulate these approaches are a focus on the involvement of end-users in the development process, and whether (a) they need the system, (b) they accept or use the system, and (c) there is an impact on outcomes from the use of the system. Despite the underlying similarity of these approaches, there is heterogeneity in the terminology of co-design approaches leading to variation regarding the stages and levels in which stakeholders should participate in the development process, as well as the specific user testing measures employed (3,4).

Barki and Hartwick authored an article in 1989, which asked us to rethink what it means to involve someone versus having someone participate. Involvement can be described as the extent to which a system reflects what is important and relevant to the stakeholder, versus participation which can be described as the extent to which specific actions or behaviors are employed by the stakeholder during system development (5). As we look at the studies

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included in this review; the term involvement has many connotations ranging from soliciting end-user insight to involving them as co-designers.

Determining the value of the stages and levels of participation is further complicated when we define this based on the target population for this systematic review, of those aged 60 or older. The World Health Organization (WHO) estimates that by 2050, the world population of those aged 60 and older will increase to 2 billion, or nearly double what it was in 2015 (6). As the number of people within this age group increases, aligning systems that meet their needs is imperative. While user needs and experience with technology may vary at any age, system designers have focused on proven areas that address the accessibility needs when designing for the aging population (7,8), but less has been done to evaluate the outcomes associated with involving older adults at various stages and levels during the development of EHTs. Involvement and participation occur at varying levels, and when users are more fully engaged, they feel a stronger sense of participation, which has the potential to impact their outcomes (2). While several factors make up the stages of involvement and participation, further evaluation as to the level of involvement at each stage is necessary, as little is known about which approaches are the most effective in engaging older adults (9). The intent to involve stakeholders can be difficult as the framework to conduct these efforts are not prolific (3). This is further exacerbated by the hierarchical challenges that may arise between designers, healthcare providers, and older adults, which can lead to selective inclusion of ideas (7,10–12). However, by addressing this challenge, the opportunity for mutual learning occurs, allowing knowledge transfer among stakeholders (13,14).

In a previous systematic review, evidence supported involving older users in technology design, as indicated in the positive consequences of involvement, specifically regarding learning, adjusted design, and an increased sense of participation (15). Further supported in the review are the comparison examples between outcomes of designs that involved older users during the development of the tool versus those that did not, and of the results of those who involved older adults the developed tool was determined to better fit the needs of the intended population (15). EHTs were selected for this review because the evidence supports the significant impact these tools have on improving health outcomes, including increasing knowledge, improvement in risk perception, and improvement in communication between patients and their clinicians (16). While evidence supports the value in involving older adults in the design process, and positive impacts have been reported in the utilization of EHTs, we need to be cognizant that inconsistencies exist within these approaches, therefore evaluating the involvement process further is necessary to ensure that the expended efforts of older adults result in a mutual exchange of knowledge and improvements to the development process without detriment to their well-being.

For co-design to work effectively, all stakeholders must be provided opportunities to equally engage, and their values and culture incorporated into the development process. Failing to

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optimize co-design approaches across the development process limits the representation and empowerment of stakeholders (2).

A previous systematic review published by Lancaster et al. focused on electronic health tools but synthesized the usability and effects (17) of the tools, rather than co-design approaches. Additionally, their target population was not defined as older adults. Another systematic review was published with similar methods and target populations as our review (1), however, they focused on assistive technology which promotes independence of living rather than on the development of EHTs. While similarities exist between our review and these two, we have enough distinction from these reviews that our search will draw from a different group of studies, therefore will provide for a robust review of the current state of co-design approaches used to involve older adults in the development of EHTs.

Aim

This systematic review synthesizes the current state of utilizing co-design approaches with older adults in the development of electronic healthcare tools, how the approaches are used, classified, including terminologies, frameworks, stages of development, involvement processes, levels of participation, and user testing measures. The types of EHTs considered for this review are those which provide patients with the ability to acquire information with the sole intent to support their healthcare decisions.

Methods

Patient and Public Involvement

No patient involved

Systematic Literature Search Strategy

We conducted a systematic review, following the Preferred Reporting Items for Systematic Reviews (PRISMA) 2020 checklist (18). Consultations with an experienced research librarian (JB) were held to develop our search strategies. In March 2021, searches were developed for PubMed, Embase, and Scopus. A broad search strategy was utilized to ensure all “collaborative design” approaches were captured in this review.

(Table 1 - Search Results)

The search results were screened according to the criteria defined in Table 2.

(Table 2 - Inclusion and Exclusion Criteria)

We are interested in the approaches used from studies that involve older adults in the development of EHTs, specifically the tools used to inform their healthcare decisions.

Data Extraction and Synthesis

Data extraction (selection and coding)

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All records were uploaded to Covidence for screening purposes (19). The authors (AC, KA) independently screened titles and abstracts that were identified in the search results from the database and excluded those which did not meet the selected inclusion criteria. A third reviewer (LM) resolved conflicts. After acquiring the full-text article, each author (AC, KA) independently reviewed and assessed each article to determine if it met the inclusion criteria. The final list was reviewed by the two authors to confirm which articles should remain in the review. A third reviewer (LM) resolved conflicts.

Study data were extracted for analysis and included study setting, population characteristics (age and health status), theoretical frameworks and principles used, terms used to define co-design approaches, types of co-design involvement processes (workshops, focus groups, interviews, prototyping, think-aloud, observation, contextual inquiry, usability testing, etc.), levels of participation, types of electronic healthcare tools, and user testing measures.

Risk of bias (quality) assessment

While our systematic review was not focused on reported outcomes, and may not have required a quality assessment, as suggested in similar systematic reviews related to involvement processes, we wanted to provide qualifications as to how we assessed each study (15,20). We evaluated the quality of studies included in this review, using the Agency for Healthcare Research and Quality (AHRQ) Evidence-based Practice Center approach (21). Per this approach, two authors (AC and KA) rated studies on directness, consistency, limitations, and reporting bias, and a consensus was reached through a discussion of conflicts (21).

Strategy for data synthesis

The data from each study were combined to provide a narrative or descriptive overview. We developed a data extraction form in Excel using a random sample of 2-3 included full texts of articles and revised it iteratively. The main characteristics from the included studies are outlined in Table 3 below. A comprehensive assessment of all study results has been conducted, highlighting similarities and differences in each study regarding the types of co-design approaches utilized. This comparison was necessary, as there was anticipated heterogeneity in the terminology and approaches. We used widely reported classification methods to categorize the included studies on the types of involvement processes (22), levels of participation (3), types of healthcare tools (23), and older adults' needs addressed by electronic healthcare tools (24). We mapped the included studies on methods used and levels of participation and cross-classified the older patient's needs addressed by electronic healthcare tools reported in each article with the type of electronic healthcare tools.

(Table 3 - Study Characteristics)

Involvement processes were extracted from the studies and classified by the participatory co-design framework originally developed by Leinonen and since further refined, representing 4 distinct phases, including contextual inquiry, participatory, design, product design, and software

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prototype as hypothesis (functional prototypes) (22,25). Levels of participation were extracted and classified based on Vaughn's framework, representing involvement levels ranging from being informed, consulted, involved, in collaboration as a co-leader, to empowering oneself and others (3). Leinonen and Vaughn's framework were combined to evaluate how the core involvement processes that span across the various co-design approaches relate to the five levels of participation, as depicted in Figure 1- Extracted from Vaughn's Involvement Framework and Leinonen's Process Framework and modified to reflect the bi-directional aspect of involvement as a process in collaboration with the stages of the co-design involvement processes.Figure 1.

(Figure 1- Extracted from Vaughn's Involvement Framework and Leinonen's Process Framework and modified to reflect the bi-directional aspect of involvement as a process in collaboration with the stages of the co-design involvement processes.)

Within Leinonen's framework, participatory design is defined as the stage at which stakeholders have the opportunity to provide input (25). For this literature review, this subcategory encompasses all involvement processes extracted from the studies in which stakeholder insight was represented. Other involvement processes were defined but did not fit within the cataloging of processes, which included the use of eye-tracking (8,26), teach-back methods (27), and spending time teaching older adults how to use technology in a classroom setting to better prepare them for engaging in the development process (14). While the latter two methods do not fit within the cataloging of processes, the extent to which this engagement occurred provided a higher level of participation suggesting that participants were empowered through these methods.

Study Selection

The initial database search returned 835 articles, including 495 from PubMed, 246 from Embase, and 94 from Scopus, of which 210 were duplicates, leaving 625 articles for the title and abstract review. Upon completion of the title and abstract review, 156 articles remained for a full-text review. The further assessment resulted in 25 studies that met the inclusion criteria. The review process is depicted in the PRISMA flow diagram (28) in Figure 2.

(Figure 2 – PRISMA Flow Diagram)

Results

Characteristics of Included Studies

Twenty-five eligible studies were remaining from our initial database search. All studies indicated that older adults were involved in the development of an EHT. We were unable to obtain which co-design approach was used in 2 of the included studies (26,29), and 2 did not define a specific theoretical framework or co-design principle (8,30). While all studies included older adults over the age of 60, we were unable to define the overall mean age, as 6 studies simply provided the age range (7,27,31–34), and 3 simply stated participants were 65 or older

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(11,12,14). Overall, the studies included participants in the age range from 60 – 91, and of the studies which provided specific ages, we have a mean age of 71.4.

Of the 25 studies, the sample size across all studies ranged from 7 – 135 participants, with a mean sample size of 35.5 for participants classified as older adults, and a mean sample size of 40.8 when including all participants.

The settings and locations varied by study. Of the 25 studies, only 11 reported a specific setting in which the study was held, including laboratories, clinics, homes, community and senior centers, as well as remote sessions via Zoom. Of the studies that met inclusion criteria, locations included Canada, Denmark, Hong Kong, Ireland, Italy, Netherlands, Sweden, and the United States of America.

All included studies were found to be of high quality, with no concerns regarding directness, consistency, precision, and reporting bias. We did find some limitations, which we rated as "medium" level, including two studies that did not indicate frameworks (8,30), and six of the twenty-five studies had limitations due to sample size (8,11,33,35–37). Overall, we did not find any studies that would rate as a "high" level limitation.

(Table 4- Summary Characteristics)

Co-Design Approaches

Definitions

Overall, 9 different co-design approaches were represented among the 25 studies, as indicated in Table 5. Of the 25 studies, 7 studies indicated the use of more than one co-design approach (8,9,32,33,38–40), and 2 studies stated that stakeholders were involved in development but did not indicate a specific approach (26,29). Of those that indicated a specific approach, 10 did not define the co-design approach which was utilized in the study (8,9,27,30–33,39,41,42). Reviewing the definitions of co-design approaches, 8 studies refer to involvement with end-user/stakeholder, 5 studies refer to system usability, 4 studies refer to user needs, 3 studies refer to prototypes, and 3 studies refer to iterative development. Interestingly, while only 3 definitions refer to prototypes and 3 refer to iterative development, prototyping and iteration appeared most frequently in the studies.

(Table 5 – Co-Design Approaches and Definition)

Iterative Development

Overall, 19 of the 25 studies conducted their study in multiple phases, ranging between 2 to 6 phases, and an overall mean of 3.4 phases per study. As reported in Table 5, 7 studies indicated the use of multiple co-design terms to define their approach, therefore iterative development is captured for each term defined by the study and reported in Table 6. Of the 6 studies which did not employ iteration in development, 2 labeled user involvement as either a human factors or

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usability evaluation approach, and as noted in the definitions table, did not provide definitions to the approach utilized (30,39).

(Table 6 – Iterative Development compared against Co-Design terminology)

Frameworks and Principles

Theoretical Framework/Co-Design Principles

Despite iterative development and prototyping not being included in the definitions of co-design approaches, studies did not necessarily determine involvement processes by how they defined the approach. Studies did indicate that specific frameworks or principles were utilized to guide them in selecting which involvement processes to follow. Studies that reference the same terminology for both the approach and framework such as Human-Centered Design (HCD), did not use iteration in their definition, but referenced standards such as HCD Methodology – ISO 9241-210 which specifically states this is an iterative process (43).

Of the 25 studies, 2 did not indicate that their study was guided by a particular framework or principle, although one study was defined as using co-design with a participatory experience, and the other as using human factors, neither provided clarification on their approaches. As 18 different frameworks or design principles were extracted from the studies, it is difficult to conclude which involvement processes should be included within each framework. However, there are inconsistencies among the studies which indicated utilizing similar frameworks or principles, such as HCD, UCD, and IPDAS, which warrants further exploration into what it means to state a study is employing a particular framework or principle.

(Table 7 – Theoretical framework and Co-Design Principles)

Stakeholder Involvement

Participant Types

Each of the 25 studies indicated that older adults were involved in the development as the intended users of the EHT. While all studies included older adults over the age of 60, we were unable to define the overall mean age, as 6 studies simply provided an age range (7,27,31–34), and 3 simply stated participants were 65 or older (11,12,14). Of the 25 studies, 13 studies engaged additional participants including caregivers, healthcare providers, and others such as (researchers (38,41), department heads & policymakers (41), and control group participants (39)). A total of 9 studies targeted participants with specific diagnoses, including COPD, larynx cancer, heart failure, hearing impairment, mild cognitive impairment, hypertension, asthma, prostatitis, and hypotension.

(Table 8 – Study Participants)

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Types of Involvement Processes

A variety of involvement processes were used in the 25 studies, as indicated in Table 9, which reports 151 processes across all the included studies. Each study utilized between 2 to 11 involvement processes, with a mean of 6 processes per study.

(Table 9 – Co-Design Involvement Processes)

When cross-classifying the involvement processes with the co-design approach used in the study, we see that multiple involvement processes are spread across the studies, except for those approaches which did not provide a clear definition, such as human-centered computing, human factors, interaction design process, and usability evaluation approach.

(Table 10 – Co-Design Approaches cross-classified with Co-Design Involvement Processes)

Levels of Participation

Defining each study based on the level of participation presented challenges. Each study was assessed to have reached a participation level based on the definitions presented in (Table 11), as provided by the research process developed by Vaughn (3). While Vaughn's definitions are related to research processes, they can be translated to the levels of participation in developing EHTs. The challenge in mapping each level against the co-design approach is that the studies did not define distinct points at which participants were involved. However, we can see that most studies employed the first three levels of participation, that of inform, consult, and involve, as evidence suggests the information provided to the research team was valued and assessed for inclusion into various design elements of the tool. However, fewer studies indicated that participants had an equal level of decision-making when making iterative changes to the design, and even fewer empowered the participants to take a lead in engaging others on the usefulness of the tool, as represented in (Table 12), where the levels of participation were cross-classified with co-design approaches.

(Table 11 - Levels of Participation)

Comparing all 25 studies to those classified as Level 5-Empower (Yes), as depicted in (Figure 3) we noted an increase in contextual inquiry (from 4% to 5%) and functional prototypes (from 10% to 14%), and a decrease in the utilization of participatory design (from 60% to 55%) and product design (from 24% to 23%). This is more distinct when comparing the utilization of processes in studies classified as Level 5-Empower (No) versus Empower (Yes), specifically regarding the decrease in participatory design (from 69% to 55%) and increase in functional prototypes (from 2% to 14%), respectively.

(Figure 3 - Involvement Processes (Percent of Total))

(Table 12 - Cross-classifying co-design approaches with levels of participation)

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Electronic Healthcare Tools (EHTs)

Types of Electronic Healthcare Tools (EHTs)

The stated aim of most of the studies was to involve older adults in gathering their insight on developing or redesigning an EHT. This review focused on studies that involve older adults in the development of EHTs, specifically the tools used to inform their healthcare decisions.

To determine whether the tool described in the study met these criteria, each tool was classified using the World Health Organization (WHO) Classification of Digital Health Interventions, which is a framework that focuses on objectives within health sectors and categorizes the digital technology that fits within each sector (23). The classification scheme focuses on interventions for 4 primary users, including clients, healthcare providers, health system or resource managers, and data services. Our target population is older adults, therefore they fit within the client framework, but it should be noted that caregivers also fit within this grouping. One of the four primary use cases for the WHO Classification framework is to assist in synthesizing evidence and should be acknowledged that the terminologies and definitions were refined by obtaining public feedback during working group sessions (23), therefore is an appropriate framework for categorizing the data extracted from each study included in this review.

The classification tool for clients consists of 7 categories, each further broken down into subcategories, as depicted in Table 13. The tools which were developed or redesigned within each included study have been classified within 5 of the 7 overarching WHO categories, including 1.1 Targeted Client Communication, 1.3 Client to Client Communication, 1.4 Personal Health Tracking, 1.5 Citizen Based Reporting, and 1.6 On-Demand Information Services to Clients.

In 12 of the 25 studies, the EHT that was developed or redesigned fits within multiple categories, therefore, was cataloged in multiple categories. Of the 25 studies, 12 developed a tool that spanned multiple categories. 16 of the 25 studies focused on tools that directly supported the ability to search for health information on a particular topic. 18 of the studies focused on health tracking, but within this category 3 were specific to accessing a personal health record, 7 focused on wearables or sensor technology for tracking purposes, and 8 focused on journaling or documenting current health status and activities. Additionally, 4 studies focused on communication of health information, and 2 focused on reporting on interactions with healthcare workers. The overall objective was to ensure the tools included in the selected studies fit within a categorization that could be defined as those which support healthcare decisions, which was achieved through this classification process.

(Table 13 –WHO Classification of Digital Health Interventions)

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Older Adults' Health Needs to be Addressed by EHTs

To categorize the older adults' needs we used the definition of Hermann et al. They reported the concept of "customer needs addressed" by the healthcare industry and we adapted these definitions to identify older adults' health needs addressed by EHTs.

(Table 14-Older adults' needs addressed by electronic healthcare tools)

Just as the EHTs spanned multiple categories of the WHO classification, as indicated in Table 13, we also see in Table 14 that these tools span multiple classifications of technology needs, with 20 studies indicating they were meeting the technology needs of patients through tools that focused on patient engagement.

We also cross-classified the older adults' needs addressed by electronic healthcare technology reported in each article with the type of electronic healthcare technology as categorized by the WHO classification of digital health interventions, as indicated in (Table 15).

(Table 15 - Cross-classification of older adults' healthcare technology needs with the WHO classification of digital health interventions)

Of the 20 studies that indicated that the EHT met the technology needs of patient engagement in their healthcare, 14 were cross-classified as an on-demand information service intended for clients to look up their health information. This further validates that each study met the inclusion criteria of providing information to older adults for the sole purpose of supporting healthcare decisions.

Subjective and Objective Measures

We extracted 19 different measures, including 13 subjective, 4 objective, and 2 physiological (eye-tracking) measures, accounting for 19 of the 25 included studies, as indicated in Table 16. These studies utilized between 1 to 6 measures, with a mean of 3.2 measures per study, with usability as the most common endpoint, being measured in 13 of the included studies. Of the 19 studies that utilized these measures, only 5 studies used objective measures but were focused on three specific endpoints (errors, performance, and need for assistance). Within the 18 studies that used subjective measures, we extracted 23 different endpoints. Of these 18 studies, 5 utilized validated measures only, 8 utilized non-validated measures only, and 5 studies used a combination of both validated and non-validated measures.

(Table 16 – User Testing – Subjective and Objective Measures)

Discussion

Key Findings

This review focused on synthesizing the current state of utilizing co-design approaches with older adults in the development of EHTs and aimed to evaluate the extent to which co-design approaches engage older adults throughout the development lifecycle. We extracted evidence

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from each study to categorize the terms used to describe co-design approaches, as well as to draw themes around how these terms were defined. This review also explored the frameworks and principles that determined the stakeholder involvement, including types of involvement processes and the levels of participation. We evaluated the EHTs and categorized them by health sector classifications, the health care needs being addressed by the EHT, and the subjective and objective measures employed.

Co-Design Approaches

As anticipated, heterogeneity exists within the definitions used to describe co-design approaches. While 9 definitions were represented across the studies, general themes presented suggest that the approaches could be used interchangeably. While this may further exacerbate a problem, and lead to inconsistent use of terminology, what is of notable importance within this review, is not so much on the approaches used, but how they are interpreted and then employed to determine what user involvement means. Not surprisingly, each study specified either definitively within their aim or subsequently, that end-users should have some level of involvement or participation during the development life cycle process. However, fewer studies indicated that participants had an equal level of decision-making when making iterative changes to the design, and even fewer empowered the participants to take a lead in engaging others on the usefulness of the tool. Of interest, is that within the studies that empowered participants, we observe an increase in the utilization of specific involvement processes such as contextual inquiry and functional prototyping.

Frameworks and Principles

We extracted 18 different frameworks or design principles from the 25 studies, and 2 of the studies did not mention a particular framework or principle. We chose not to cross-classify this variable as to not misrepresent relationships; however, this diversity exemplifies a need for a framework that is specific to engaging older adults in the development of EHTs, which can be consistently utilized, and which also provides validation to the involvement processes that should be employed.

Stakeholder Involvement

While we found inconsistency in the involvement processes among the studies, over 70% included interviews and prototyping, others included a range of processes such as cognitive walkthroughs, think-aloud sessions, surveys, scenarios, and focus groups, and only 20% of the studies utilized contextual inquiries and use cases. We also found inconsistent use of designing the study in phases, however, 19 of the 25 studies reported the value of iterative development and averaged 3.4 phases per study. It should be noted that when working with an older adult population, specifically when working with a population with a severe diagnosis, extending development over multiple sessions can present additional challenges, as their needs and abilities may affect their ability for involvement (1).

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To the best of our knowledge, current frameworks, or principles have not provided guidance on distinguishing the levels of participation, nor the validated involvement processes that should be utilized when engaging older adults in developing EHTs. To provide structure to this notion, this review drew upon Leinonen's framework for processes and Vaughn's framework for levels of participation. After extracting evidence of both variables from the 25 studies, and overlaying these on top of the two frameworks, we began to recognize the areas in which we need to further explore. Specifically, if the purpose is to involve end-users in the development process, we must be cognizant of the end goal, and whether that is to develop a tool that will be useful and utilized, or more importantly, to develop a tool that through the involvement process also can bring a sense of empowerment to the end-user.

It has been suggested that exposure to the technology before the usability testing has shown to be an effective means of eliciting users' needs, such that the increased knowledge and awareness of the tool inspires ideation and innovation (9,33), and that tacit knowledge or real-life context increases the ability to portray specific user needs (41). However, there are concerns about bias if an individual is exposed to technology before soliciting their design ideas (44). Not to dismiss the concern for bias, but further evidence supports the notion that experienced users of technology can provide more meaningful insight during design sessions and could provide feedback more representative of the target population (45). We also see that allowing users to engage with the technology for an extended time or ensuring they understand how to use the technology, reduces frustration when contributing to design ideas (14,27,44). For example, when participants were asked to participate in two similar design activities, they were more engaged when they had familiarity with the topic, compared to being asked to innovate on a novel topic (35).

Suggestions have been made to separate co-design processes from outcomes and focus on the role that participants have to be co-designers and co-initiators (46), which supports the values defined by the fourth and fifth levels of involvement, that of collaborate and empower, respectively. We should be looking at involvement as a process, one in which through various stages of the development life cycle, users have the potential to cross through the spectrum of involvement highlighted in Vaughn's framework. By combining Leinonen and Vaughn's framework, we can see how the involvement processes that span across the various co-design approaches, can be used in iterative development, but through these various stages, users can engage through the process of being informed, consulted, involved, in collaboration or as a co-leader, to empowering oneself and others. Further, we suggest that additional exploration as to the extent to which a greater understanding of the tool can contribute to richer feedback (9), therefore suggesting the levels of participation are bi-directional, as depicted in Figure 1, and when realized within the development lifecycle will result in greater contributions to the development of the tool.

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Electronic Healthcare Tools (EHTs)

While the EHTs that were applied in the studies spanned multiple categories under the WHO classification, the majority focused on tools that provided the ability to search for information on a particular health topic or that allowed participants to track their health information through self-monitoring, including accessing personal health records, use of wearables, or documenting health information. Through further cross-classification, we indicated that most tools in the included studies were focused on patient engagement. If the intent is to develop a tool focused on engaging patients in their healthcare decisions, and positive impacts have been reported in the utilization of electronic healthcare tools, and improvements to the tools are supported through appropriate involvement, then further evaluation needs to occur to ensure older adults are involved at the appropriate stages and levels to create the flow of bidirectional knowledge.

Furthering the notion that a framework, specific to older adults in developing EHTs, needs to be evaluated, is concerning the inconsistency of user testing assessments. We extracted evidence that 19 different subjective and objective measures were utilized in 19 of the 25 included studies. We further categorized these measures to evaluate whether studies were using validated measures and found that only 5 studies (20% of all included studies) used validated subjective measures to evaluate their EHT. Other than the SUS scale, which was utilized in 20% of the studies to measure usability, the other validated subjective measures identified were only used in 4% - 8% of studies, while 12 distinct non-validated Likert-scale assessments spanned across 52% of all studies. Usability was also the most specified measure in non-validated Likert-scale assessments, taking place in 8 studies. Overall, 13 of the 25 studies reported measuring usability, but less than half used a validated measure. This discrepancy in assessments for user testing makes it difficult to draw definitive conclusions as to whether studies are consistently measuring what they intended to measure.

Limitations

As our inclusion criteria spanned an array of terms that we categorized as co-design approaches, we had several approaches that were only represented by one study, which may not fully represent all studies that incorporate such approaches. In areas in which we extracted the levels of participation, specific evidence was difficult to draw out, as to which levels of engagement were employed at specific processes, therefore generalizing this information was limited. Evidence was also extracted to classify the EHTs, but as each tool did not fit precisely within the WHO classification or Hermann's technology needs classification, it was difficult to ensure each tool was represented accurately.

Conclusions

This review depicted not only the heterogeneity in the terminology used to define co-design approaches, but the plethora of frameworks and principles utilized when stating co-design approaches are being applied in developing electronic healthcare tools for older adults. Most

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studies in this review developed EHTs specific to patient engagement and designed their study to include older adults' feedback in the design, indicating a collaborative focus centered on meeting the needs of this population. While evidence extracted from included studies suggests that co-design approaches aim to involve end-users in the iterative development of healthcare tools, there are gaps in the evidence to support the levels at which users should be involved and the extent to which this involvement can increase the user's knowledge of the technology. Evidence extracted from the studies suggests that the involvement processes utilized may be related to the levels at which stakeholders participate and can either limit or encourage collaboration and empowerment of stakeholders. Evidence also suggests that the utilization of validated assessments for user testing was not consistently utilized. To ensure each of these factors is optimized, additional research is needed to evaluate the feasibility of developing a co-design framework specifically for developing EHTs for older adults.

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Contributors

AC and KA conceived of the broad concept for the literature review. AC developed the protocol, then worked with KA and JB to develop a search strategy. AC and KA independently screened titles and abstracts that were identified in the search and excluded those which did not meet the selected inclusion criteria. LM resolved conflicts. After acquiring the full-text articles, AC and KA independently reviewed and assessed each article to determine if it met the inclusion criteria. The final list was reviewed by the AC and KA to confirm which articles should remain in the review. A third reviewer LM resolved conflicts. AC extracted and synthesized the data. AC wrote the review, with meaningful insight in the structure of the analysis and conclusion with KA and LM. Final review of the manuscript by KA, LM, AK, and DR.

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Competing Interests

None declared

Patient consent for publication

Note required

Ethics Statement

This is a systematic review, without human involvement, as such an ethical assessment was not required.

Data availability statement

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All data relevant to the study are included in the article or uploaded as supplemental information.

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Figures

Figure 1- Extracted from Vaughn's Involvement Framework and Leinonen's Process Framework and modified to reflect the bi-directional aspect of involvement as a process in collaboration with the stages of the co-design involvement processes.

Figure 2 – PRISMA Flow Diagram

Figure 3 - Involvement Processes (Percent of Total)

Tables

Table 1 - Search Results

PubMed – 3.17.2021		
Search #	Query	Results
#1	(Co-Design[tiab] OR Co-Creation[tiab] OR Co-Production[tiab] OR Participatory Design[tiab] OR Human Factors[tiab] OR User-Centered Design[mesh] OR User-Centered Design[tiab] OR Participatory Approach[tiab] OR Human-Centered Design[tiab] OR Universal design[mesh] OR Patient Participation[mesh] OR patient participation[tiab])	38,538
#2	(Aged[mesh] OR Older Adults[tiab] OR Older People[tiab] OR Elderly[tiab] OR Geriatric[tiab] OR Aged over 60[tiab] OR aged 60[tiab] OR Seniors[tiab] OR Over 65[tiab])	3,333,143
#3	(Electronic Health Records[mesh] OR Electronic Health Records[tiab] OR Electronic Health Record[tiab] OR Electronic Healthcare[tiab] OR eHealth[tiab] OR "mHealth"[tiab] OR Decision Aid[tiab] OR decision aids[tiab] OR Electronic Healthcare Aid[tiab] OR Healthcare Application[tiab] OR digital health[tiab] OR health information technology[tiab] OR health information technologies[tiab] OR health technology[tiab] OR health technologies[tiab])	59,271
#4	#1 AND #2 AND #3	495

- records before duplicates removed: 835
- records after duplicates removed: 625

Table 2 - Inclusion and Exclusion Criteria

Selection Criteria	*Inclusion Criteria*	*Exclusion Criteria*
Participants	Older adults aged 60 or older, without regard to gender, race, or ethnicity	If the study population included adults of all ages, including greater than 60 years of age, but evidence could not be extracted from those greater than 60 years of age, the study was excluded

Study Setting	All study settings were included	
Study Design	All study designs were included	
Methods and Tools	Studies published that employed co-design approaches in the development of electronic healthcare tools	Studies that used non-electronic healthcare tools
Language		Studies in which publications are not in the English language were excluded

Table 3 - Study Characteristics

Category	***Variable***	***Classification method***
Summary characteristics	Year of publication	Metadata
	First author's geolocation	Metadata
	Study setting	Metadata
	Sample size	Metadata, mean
	Number of older adults	Metadata, mean
	Study population	Metadata
Co-design approaches	Terminology or definitions	Manual, binary
	Theoretical framework/design principles	Manual, binary
	Iterative development	Metadata, mean
Stakeholder involvement	Participant types	Metadata
	Types of Involvement Processes	Leinonen et.al (22)
	Levels of participation	Vaughn et.al (3)
Electronic healthcare tools	Types of healthcare tools	WHO classification of digital health interventions (23)
	Older adults' needs addressed by electronic healthcare tools	Hermann et.al (24)
	User testing	Metadata

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Table 4- Summary Characteristics

First Author's last name	Year of publication	First author's geolocation	Study setting	Sample size	No. of Older Adults	Study population				Age of older adults (mean)	Diagnosis
						Older Adults/ Patients	Caregivers	Healthcare Providers	Others		
Abujarad	2021	USA	Yale School of Medicine	38	24	X	X	X		69	
Ahmed	2019	USA		9	7	X	X			67	Heart Failure
Bogza	2020	Canada	Clinics, research center, or in their own homes.	19	12	X	X	X	X	(2) 60 – 64 years (6) 65 – 74 years, (3) 75 – 84 years, (1) >85 years	Mild Cognitive Impairment (MCI)
Cornet	2020	USA	Laboratory	63	50	X	X	X	X	>65	Heart Failure
Gonzalez	2014	USA		12	12	X				77.7	
Grimaldi	2020	Italy		10	10	X				> 60	
Gustafson	2016	USA	Home visits	135	135	X	X	X	X	> 65	
Harrington	2018	USA		39	39	X				72.1	
Harte	2017	Ireland	Participant's home	12	12	X				61-85	
Hoffman	2020	USA		16	8	X	X			Phase 2: 65-84 Phase 3: 61-89	
Holden	2020	USA	Eskenazi Health	23	23	X				67.6	
Kim	2018	USA	Participant's home or senior center	17	17	X				65-80	
Mansson	2020	Sweden		10		X				76	
Martin-Hammond	2015	USA	On-site 2 Locations	7	7	X				70	
Nguyen	2019	Netherlands		56	56	X			X	68.9	
Nielsen	2018	Denmark	Eriksholm Research Centre	54	36	X	X	X		70.3	Hearing impairment
Or	2012	Hong Kong	Community health service center	57	50	X			X	71.6	One of the following: hypertension, diabetes, heart disease, asthma,

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											prostatitis, hypotension
Petersen	2019	Netherlands		71	32	X		X		70	Larynx cancer
Poirier	2019	USA		45	45	X				72.2	
Portz	2020	USA	Home, clinic, or zoom	81	31	X	X	X		66-91	Heart Failure (palliative care patients)
Scandurra	2013	Sweden		8	8	X				65-80	
Ummels	2020	Netherlands		63	63	X			X	70.3	
Van De Dijk	2013	Netherlands		86	86	X				72.5	COPD
Van Velsen	2019	Netherlands		82	82	X	X			71.5	
Wikberg-Nilsson	2018	Sweden		7	7	X				75	Reduced hearing, eyesight, mobility, sensibility, and loss of memory function

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Table 5 – Co-Design Approaches and Definition

Co-Design Approach	Total studies (n=25), n(%)	Definition Provided	References
Co-creation	2 (8%)	1	(36,38)
Co-design	5 (20%)	1	(8,9,32,40,41)
Human-centered computing	1 (4%)	-	(9)
Human-Centered Design (HCD)	3 (12%)	3	(7,11,34)
Human Factors	1 (4%)	-	(30)
Interaction Design Process	1 (4%)	-	(9)
Participatory Design	7 (28%)	2	(8,9,31,33,35,40,42)
Usability Evaluation Approach	1 (4%)	-	(39)
User-Centered Design/Approach	9 (36%)	7	(12,14,27,32,33,37,47–49)
Total	**31**	**14**	

Table 6 – Iterative Development compared against Co-Design terminology

Co-Design Term	Studies that employed Iterative Development	No. of Phases (Mean)
Co-creation (n=2)	2	3
Co-design (n=5)	5	3.2
Human centered computing (n=1)	1	2
Human Centered Design (HCD) (n=3)	3	3.3
Human Factors (n=1)	-	-
Interaction Design Process (n=1)	1	2
Participatory Design (n=7)	5	2.8
Usability Evaluation Approach (n=1)	-	-
User Centered Design/Approach (n=9)	7	3.9
Other (Did not Specify) (n=2)	1	3
Total studies (n=25), n(%)	19 (76%)	3.4

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Table 7 – Theoretical framework and Co-Design Principles

Theoretical framework/Co-Design Principles	Total studies (n=25), n(%)	References
Age-Related Research-Based Usability Guidelines	1 (4%)	(48)
Center for eHealth and Wellbeing Research guidelines	1 (4%)	(27)
Double Diamond Method	1 (4%)	(38)
Experience-based co-design	1 (4%)	(9)
Grounded Theory Approach	2 (8%)	(31,34)
Human Centered Design Methodology (ISO Standards)	5 (20%)	(7,11,33,34,40)
Human factors design principles	2 (8%)	(39,48)
Inductive thematic qualitative analysis approach	1 (4%)	(27)
International Patient Decision Aid Standards (IPDAS)	4 (16%)	(26,27,29,32)
Karagianni's Optimized Honeycomb model	1 (4%)	(36)
Medical Research Council framework	1 (4%)	(41)
MoSCoW Method	2 (8%)	(38,42)
NIATx model as a framework	1 (4%)	(14)
Nielsen's Heuristic Principles	2 (8%)	(37,39)
Ottawa Decision Support	1 (4%)	(32)
Systematic usability study guideline	1 (4%)	(39)
Ummels Framework	1 (4%)	(38)
User Centered Design Approach	3 (12%)	(12,50,51)
None reported	2 (8%)	(8,30)

Table 8 – Study Participants

Study Population	Total studies (n=25), n(%)	References
Older Adults/Patients	25 (100%)	All
Caregivers	9 (36%)	(12,14,27,32,34,35,40,42,50)
Healthcare Providers	7 (28%)	(12,14,27,29,34,40,50)
Additional Participants	6 (24%)	(12,14,27,38,39,41)
Targeted Diagnosis	9 (36%)	(8,12,27,29,34,35,39,40,51)

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Table 9 – Co-Design Involvement Processes

■ Contextual Inquiry ■ Participatory Design ■ Product Design ■ Prototype ■ Other

(color-coded to reflect association in Figure 3)

	Involvement Processes	Total studies (n=25), n(%)	References
Contextual Inquiry	Contextual Inquiry	6 (24%)	(9,14,31,35,50,52)
Participatory Design	Survey	10 (40%)	(7,9,11,12,31,32,35,47–49)
	Interview	18 (72%)	(7–9,11,12,14,27,30–32,37–39,41,48,49)
	Cognitive walkthrough	8 (32%)	(7,11,12,14,27,32,37,49)
	Think aloud	11 (44%)	(11,12,27,29,32,34,36,39,41,48,49)
	Remote evaluation	1 (4%)	(38)
	Focus group	8 (32%)	(9,14,31,32,35,38,40,47)
	Task analysis	10 (40%)	(7,11,12,26,36,38,39,41,48,49)
	Scenarios	7 (28%)	(7,12,26,32,37,42,47)
	Prototype (low fidelity)	8 (32%)	(7,14,31,32,35,37,40,42)
	Working Group Workshop	9 (36%)	(8,9,31,33,35,36,41,42,47)
Product Design	Use Cases	4 (16%)	(7,11,12,53)
	User Stories	9 (36%)	(14,27,31,32,40–42,48,49)
	Prototype (medium fidelity/throw-away)	18 (72%)	(7,8,11,12,14,27,29,30,32,35–41,48,49)
	Heuristic evaluation	5 (20%)	(7,12,32,37,39)
Software Prototype as Hypothesis	Prototype (functional/Beta version)	15 (60%)	(7,9,11,14,27,29,32,33,35,36,38,40,41,48,49)
Other	Other	4 (16%)	(8,14,26,27)
Total		**151**	

Table 10 – Co-Design Approaches cross-classified with Co-Design Involvement Processes (color-coded to reflect association in Figure 3)

Co-Design Approach / Involvement Process	Co-creation (n=2)	Co-design (n=5)	Human centered computing (n=1)	Human Centered Design (HCD) (n=3)	Human Factors (n=1)	Interaction Design Process (n=1)	Participatory Design (n=7)	Usability Evaluation Approach (n=1)	User Centered Design/Approach (n=9)	Other (Did not Specify) (n=2)
Contextual Inquiry		1	1	1		1	3		2	
Survey		2	1	2		1	3		5	
Interview	1	4	1	3	1	1	2	1	8	1
Cognitive walkthrough		1		2					6	
Think aloud	1	2		2				1	5	1
Remote evaluation	1								1	
Focus group	1	3	1			1	4		4	
Task analysis	2	1		2				1	4	1
Scenarios		1		1			1		4	1
Prototype (low fidelity)		2		1			4		3	
Working Group Workshop	1	3	1			1	5		1	
Use Cases				2					2	
User Stories		3					3		5	
Prototype (med. fidelity/ throw-away)	2	4		2	1		2	1	8	1
Heuristic evaluation		1		1				1	3	
Prototype (functional/ Beta version)	2	4	1	2		1	4		6	1
Other		1					1	1	2	

Contextual Inquiry Participatory Design Product Design Prototype Other

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Table 11 - Levels of Participation

Levels of Participation	Total studies	Definitions-Vaughn et al. (3)
Inform	25	Information is shared with community members and could become more participatory if members ask for information relative to their interests.
Consult	25	Participants provide feedback to the researchers, as related to a specific decision point.
Involve	24	Participants provide feedback to researchers during the entire process.
Collaborate	16	Participant input is valued equally, as a co-leader in the development process.
Empower	15	Participants take a key role in leading the initiative to get others engaged.

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Table 12 - Cross-classifying co-design approaches with levels of participation

Co-Design Approach	Inform		Consult		Involve		Collaborate		Empower	
	n	Ref	n	ref	n	ref	n	ref	n	ref
Co-creation (n=2)	**2**	(36,38)	**2**	(36,38)	**2**	(36,38)	**2**	(36,38)	**2**	(36,38)
Co-design (n=5)	**5**	(8,9,32,40,41)	**5**	(8,9,32,40,41)	**5**	(8,9,32,40,41)	**5**	(8,9,32,40,41)	**5**	(8,9,32,40,41)
Human centered computing (n=1)	**1**	(9)	**1**	(9)	**1**	(9)	**1**	(9)	**1**	(9)
Human Centered Design (HCD) (n=3)	**3**	(7,11,34)	**3**	(7,11,34)	**3**	(7,11,34)	**2**	(7,11)	**1**	(7)
Human Factors (n=1)	**1**	(30)	**1**	(30)	**1**	(30)	**0**		**0**	
Interaction Design Process (n=1)	**1**	(9)	**1**	(9)	**1**	(9)	**1**	(9)	**1**	(9)
Participatory Design (n=7)	**7**	(8,9,31,33,35,40,42)	**7**	(8,9,31,33,35,40,42)	**7**	(8,9,31,33,35,40,42)	**5**	(8,9,33,35,40)	**5**	(8,9,33,35,40)
Usability Evaluation Approach (n=1)	**1**	(39)	**1**	(39)	**1**	(39)	**1**	(39)	**0**	
User Centered Design/Approach (n=9)	**9**	(12,14,27,32,37,38,47–49)	**9**	(12,14,27,32,37,38,47–49)	**9**	(12,14,27,32,37,38,47–49)	**5**	(14,27,32,38,48)	**6**	(14,27,32,38,48,49)
Other (Did not Specify) (n=2)	**2**	(26,29)	**2**	(26,29)	**1**	(29)	**1**	(29)	**1**	(29)

Table 13 –WHO Classification of Digital Health Interventions

WHO Classification of Digital Health Interventions	Total studies (n=25), n(%)	References
1.1 Targeted Client Communication		
1.1.2 Transmit targeted health information to client(s)	1 (4%)	(40)
1.1.3 Transmit targeted alerts and reminders	2 (8%)	(33,39)
1.1.4 Transmit diagnostics result, or availability	1 (4%)	(35)
1.3 Client to Client Communication		
1.3.1 Peer Group for Clients	3 (12%)	(14,33,40)
1.4 Personal Health Tracking		
1.4.1 Access by client to own medical records	3 (12%)	(31,35,36)
1.4.2 Self-monitoring of health or diagnostic data by client	7 (28%)	(7,30,35,38,40,47,49)
1.4.3 Active data capture/documentation by client	8 (32%)	(9,12,27,32,34–36,39)
1.5 Citizen Based Reporting		
1.5.1 Reporting of health system feedback by clients	2 (8%)	(42,49)
1.6 On-Demand Information Services to Clients		
1.6.1 Client Look-up of health information	16 (64%)	(8,11,12,26,27,29,31–34,37,40–42,48,49)

Table 14-Older adults' needs addressed by electronic healthcare tools

Older adults' needs	Total number of studies	References
Diagnosis	0	-
Prevention	4	(36,37,48,49)
Adherence	7	(9,12,30,35,37,39,48)
Treatment	8	(26,27,29,31,32,34,38,40)
Lifestyle	11	(8,9,12,14,33,35,36,38–40,42)
Patient engagement	20	(8,9,11,12,26,27,29,31–36,38–42,47,50)
Surveillance	6	(7,12,30,35,36,38)

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Table 15 - Cross-classification of older adults' healthcare technology needs with the WHO classification of digital health interventions

Type of technology	Surveillance		Prevention		Adherence		Treatment		Lifestyle		Patient engagement	
	n	Refs	n	Refs	n	Refs	n	Refs	n	Refs	n	Refs
1.1.2 Transmit targeted health information to client(s)									1	(40)	1	(40)
1.1.3 Transmit targeted alerts and reminders					1	(39)			1	(33)	1	(33)
1.1.4 Transmit diagnostics result, or availability	1	(35)										
1.3.1 Peer Group For Clients									2	(33,40)	2	(33,40)
1.4.1 Access by client to own medical records					1	(35)	1	(31)	1	(36)	2	(31,36)
1.4.2 Self-monitoring of health or diagnostic data by client	3	(30,38,52)			1	(30)	1	(38)	3	(35,38,40)	4	(35,38,40,47)
1.4.3 Active data capture/documentation by client	2	(12,36)	1	(36)	2	(9,39)			3	(9,35,39)	6	(9,27,32,34,35,39)
1.5.1 Reporting of health system feedback by clients			1	(50)					1	(42)		
1.6.1 Client Look-up of health information			2	(37,48)	3	(12,37,48)	6	(26,27,29,32,34,40)	5	(8,12,14,33,40)	14	(8,11,12,26,27,29,31–34,40–42,49)

Table 16 – User Testing – Subjective and Objective Measures

Measures	Endpoint	Total studies (n=25) n (%)	References
Validated			
Subjective Measures			
ACE (Altarum Consumer Engagement)	Patient engagement	1 (4%)	(35)
ASQ	Efficiency	1 (4%)	(7)
CES	Confidence & Computer Efficacy	1 (4%)	(49)
Decisional Conflict Scale (values clarity)	Values Clarity	2 (8%)	(26,27)
I-PANAS-SF	Emotional Reactions	1 (4%)	(49)
Likert scale (specified validity)	Comprehension/Recall, Graphicacy, Numeracy	1 (4%)	(26)
NASA-TLX	Cognitive Workload	2 (8%)	(7,12)
NVS	Health Literacy	2 (8%)	(12,35)
Ottawa Acceptability Scale	Quality	2 (8%)	(27,32)
SURE scale	Values Clarity	1 (4%)	(32)
SUS	Usability	5 (20%)	(7,12,27,48, 49)
Us.E 2.0	Quality	1 (4%)	(11)
Non-Validated			
Subjective Measures			
Likert scale (non-specified validity)[a]		13 (52%)	(7,9,11,12,14,27,29,30,33,38–41)
Objective Measures			
Mistakes Counted	Errors	3 (16%)	(7,39,48)
Time to Complete	Performance	4 (12%)	(7,26,39,48)
Times Request Assistance	Need for Assistance	2 (8%)	(39,48)
Physiological Measures (eye-tracking)			
Cognitive Proces	Usability	1 (4%)	(8,26)
Gaze Fixation	Usability	1 (4%)	(8,26)

Satisfaction, User Acceptance, Appearance, Comfort, Functionality, Reliability, Usability, Usefulness, Ease of Use, Comprehensibility, Effectiveness, Value of Information [a]

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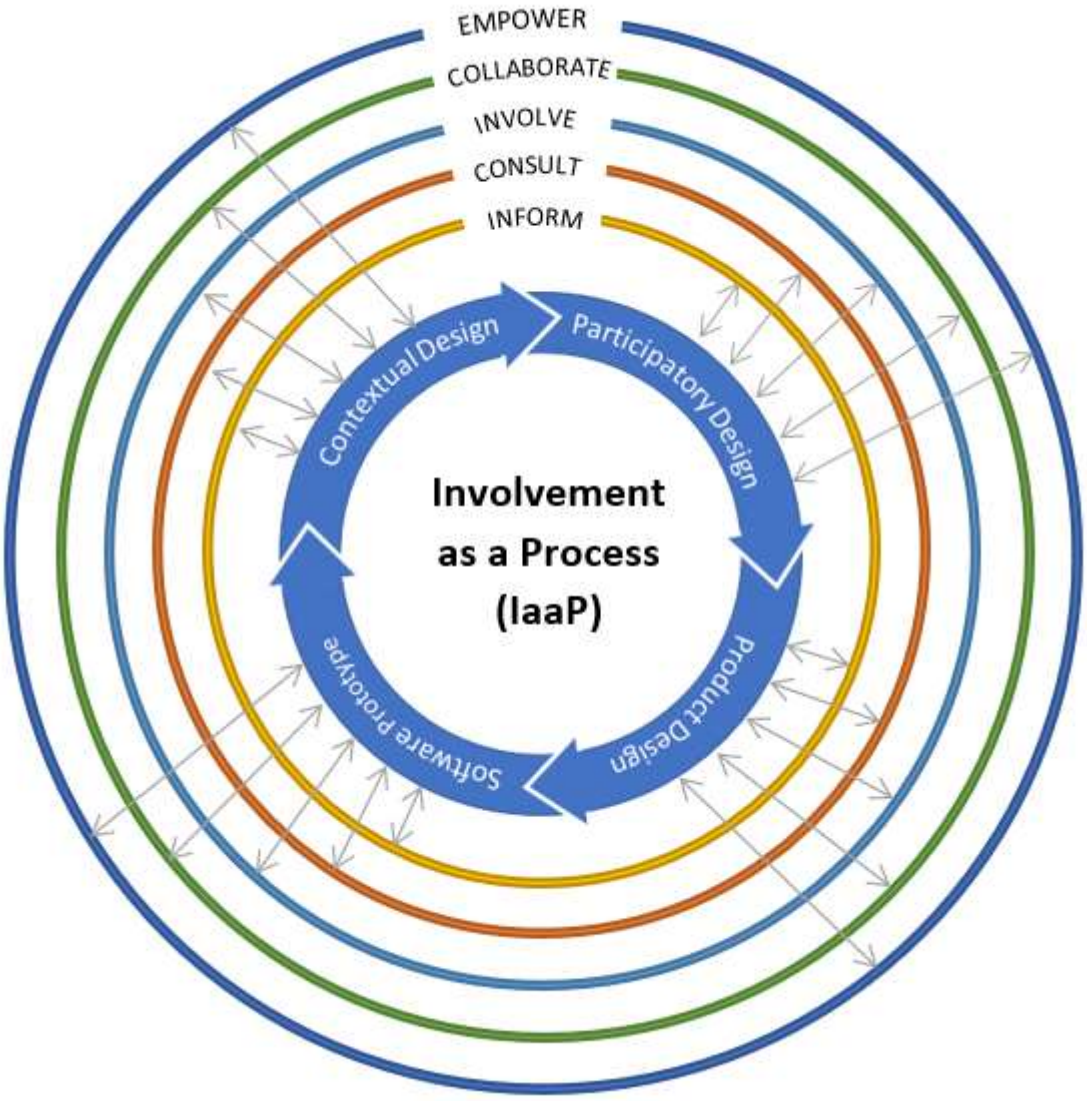


Figure 1- Extracted from Vaughn's Involvement Framework and directional aspect of involvement as a process in collabor

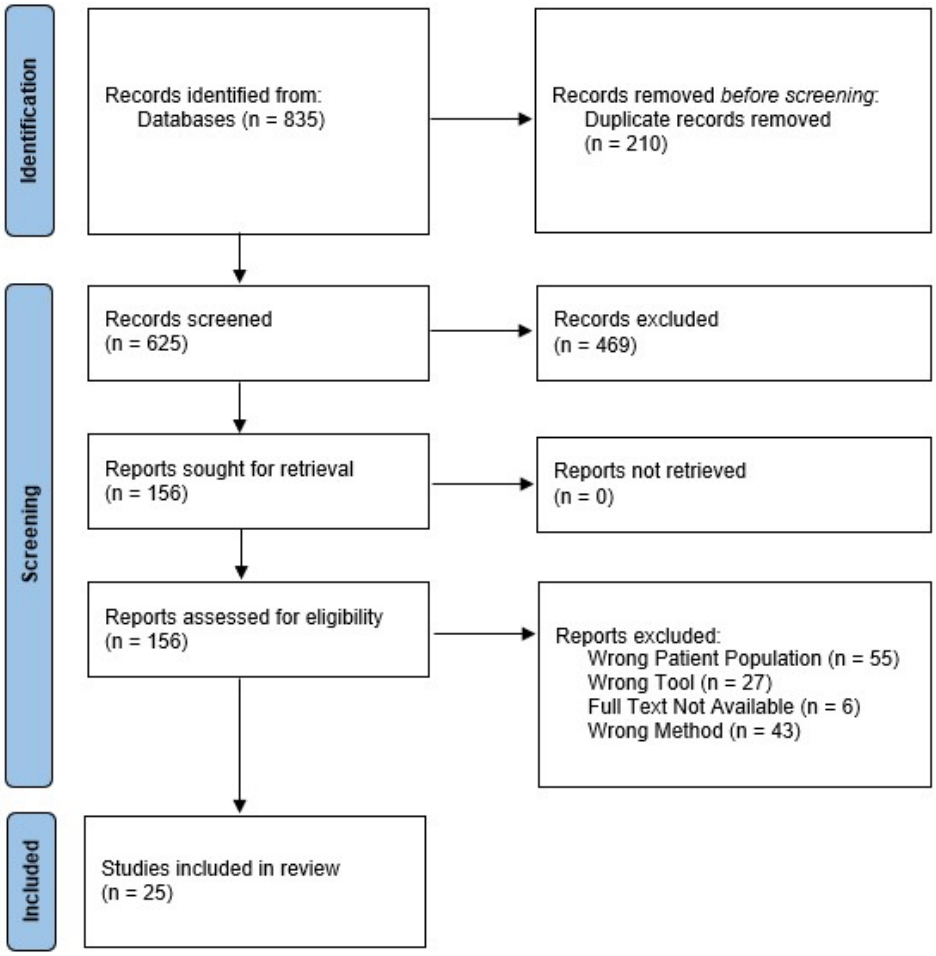


Figure 2 – PRISMA Flow Diagram

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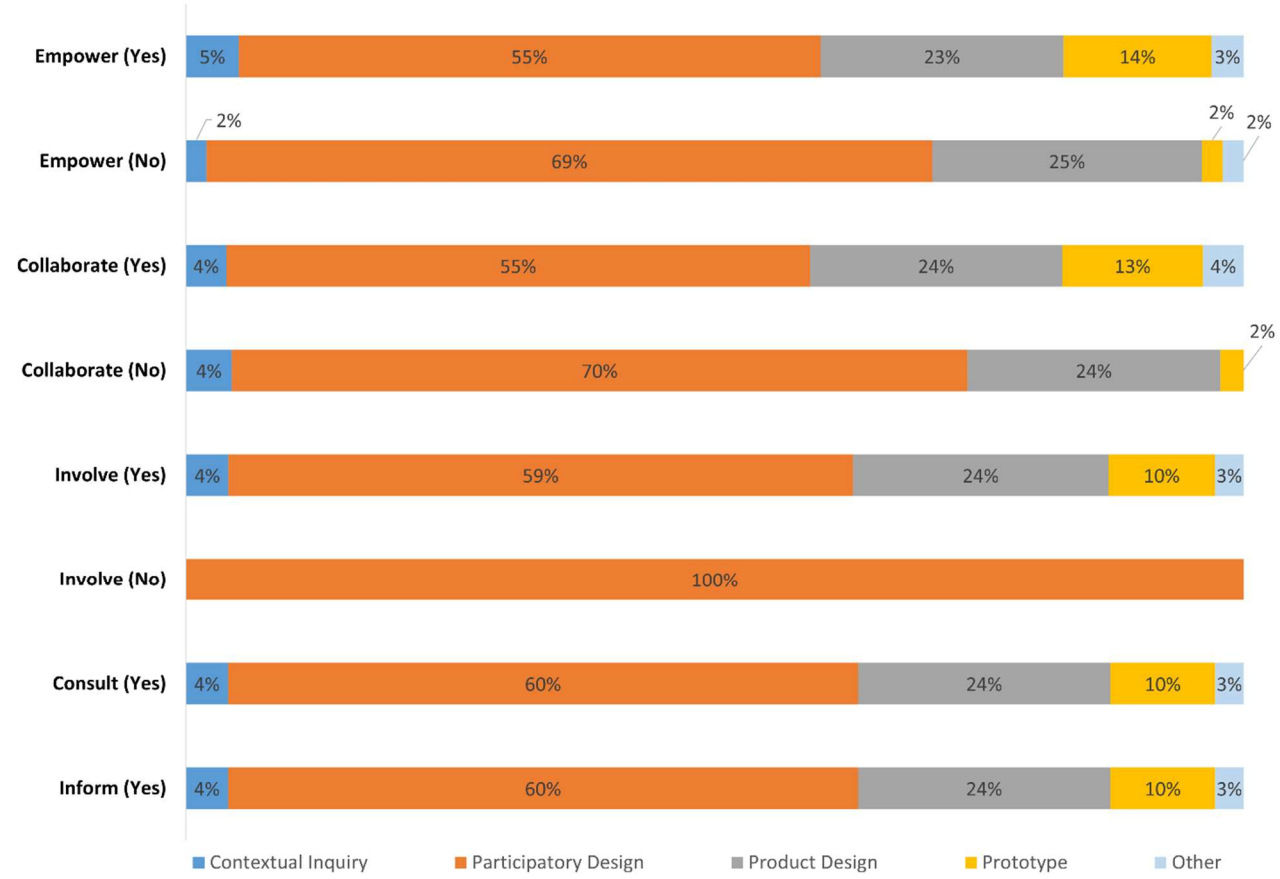


Figure 3 - Involvement Processes (Percent of Total)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2-4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	4-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	5-6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	4-6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	5
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	NA
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	4-6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	4-6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	5
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	NA
Study characteristics	17	Cite each included study and present its characteristics.	6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	8
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	6-11
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	NA
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	11-13
	23b	Discuss any limitations of the evidence included in the review.	14
	23c	Discuss any limitations of the review processes used.	14
	23d	Discuss implications of the results for practice, policy, and future research.	14
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	15
Competing interests	26	Declare any competing interests of review authors.	15
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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Co-design approaches involving older adults in the development of electronic healthcare tools: a systematic review

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Co-design approaches involving older adults in the development of electronic healthcare tools: a systematic review

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Abstract

Objective

The primary aim was to review and synthesize the current evidence of how older adults are involved in co-design approaches to develop electronic healthcare tools (EHTs). The secondary aim was to identify how the co-design approaches utilized mutual learning techniques to benefit older adult participants.

Design

Systematic review following the Preferred Reporting Items for Systematic Reviews (PRISMA) 2020 checklist.

Data Sources

PubMed, Embase, and Scopus databases were searched for studies from January 2010 to March 2021.

Eligibility Criteria

Inclusion criteria were studies employing co-design approaches to develop an EHTs, and the study population was aged 60 years and older.

Data Extraction and Synthesis

Data were extracted for analysis and risk of bias. We evaluated the quality of studies using the Agency for Healthcare Research and Quality (AHRQ) Evidence-based Practice Center approach.

Results

Twenty-five studies met the inclusion criteria for this review. All studies utilized at least 2 involvement processes, with interviews and prototypes used most frequently. Through cross-classification, we found an increased utilization of functional prototypes in studies reaching the "empower" level of participation and found that studies which benefitted from mutual learning

had a higher utilization of specific involvement processes such as focus groups and functional prototyping.

Conclusions

We found gaps to support which involvement processes, participation levels and learning models should be employed when co-designing with older adults. This is important because higher levels of participation may increase the user's knowledge of technology, enhance learning, and empower participants. To ensure studies optimize processes, participation and learning of older adults when developing EHTs, improving co-design frameworks should be evaluated.

Systematic Review Registration

PROSPERO CRD42021240013

Strengths and limitations of this study

- This review provides a comprehensive assessment of the utilization of co-design approaches in studies that involve older adults in developing electronic healthcare tools.
- We developed a broad search strategy to ensure all co-design or "collaborative" approaches were captured in this review.
- We utilized widely reported classification methods to categorize the included studies.
- We limited inclusion criteria to studies that involved older adults in the development of an electronic healthcare tool (EHT).
- We limited our analysis to the overall involvement processes and levels of participation at the study level; cross-classification of these variables at the design stage level for each study was not conducted.

Introduction

Co-design approaches aim to elicit ideas and foster a non-hierarchical environment for stakeholders, including the target population, in which their concepts, tacit knowledge, and lived experiences can be applied to develop tools that meet their needs, improve usability and impact outcomes from use of the system [1,2]. Co-design is also likened to participatory design, each defined as focusing on the engagement and creativity of stakeholders, or those untrained in design efforts, but through collaborative efforts bring mutual learning into the development process [3,4]. Despite the underlying similarity of these approaches, there is heterogeneity in the terminology of co-design approaches, which leads to variation in the evaluation measures employed, involvement processes, levels of participation, and learning outcomes [3,4]. For co-design to work effectively, all stakeholders must be provided opportunities to equally engage and learn, and their values and culture must be incorporated into the development process. Failing to optimize co-design approaches across the development process inhibits the representation, learning and empowerment of stakeholders [2].

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Barki and Hartwick (1989) suggested that we need to rethink what it means to involve someone versus having someone participate. Involvement can be described as the extent to which a system reflects what is important and relevant to the stakeholder, versus participation which can be described as the extent to which specific actions or behaviors are employed by the stakeholder during system development [5]. The term involvement has many connotations ranging from soliciting end-user insight to involving them as co-designers.

Determining the value of the stages and levels of participation is further complicated when we define this based on the target population for this systematic review of those aged 60 or older. The World Health Organization (WHO) estimates that by 2050, the world population of those aged 60 and older will increase to 2 billion, or nearly double what it was in 2015 [6]. As the number of people within this age group increases, aligning systems that meet their needs is imperative. While user needs and experience with technology may vary at any age, system designers have focused on proven areas that address the accessibility needs when designing for the aging population [7,8], but less has been done to evaluate the learning outcomes associated with involving older adults as well as the levels of participation during the development of electronic healthcare tools (EHTs). Involvement and participation occur at varying levels, and when users are more fully engaged, they feel a stronger sense of participation, which has the potential to impact their learning outcomes [2]. While several factors make up the stages of involvement and participation, further evaluation as to the level of involvement at each stage is necessary, as little is known about which approaches are the most effective in involving older adults [9]. The intent to involve stakeholders can be difficult as the framework to conduct these efforts are not prolific [3]. This is further exacerbated by the hierarchical challenges that may arise between designers, healthcare providers, and older adults, which can lead to selective inclusion of ideas [7,10–12]. However, by addressing this challenge, the opportunity for mutual learning and knowledge transfer among stakeholders can occur [2,13–15].

EHTs were selected for this review because the evidence supports the significant impact these tools have on improving health outcomes, including increasing knowledge, improvement in risk perception, and improvement in communication between patients and their clinicians [16]. We are using the term EHT to encapsulate all terms classified and defined by the WHO Digital Health Interventions [17]. EHTs are interactive web applications, mobile applications or wearables that track or provide guidance on relevant healthcare issues to improve wellbeing. For instance, monitoring health risks (i.e., risk of falling, risk of heart failure), or personal health information management (i.e., patient decision aids, access to health records and educational materials). While evidence supports the value in involving older adults in the design process, and positive impacts have been reported in the utilization of EHTs, we need to be cognizant that inconsistencies exist within these approaches, therefore evaluating the involvement process further is necessary to ensure that the expended efforts of older adults result in a mutual exchange of knowledge and improvements to the development process without detriment to their well-being.

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In a previous systematic review by Fischer et al, evidence supported involving older users in technology design, as indicated in the positive outcomes of involvement, specifically regarding learning, adjusted design, and an increased sense of participation [18]. Further supported in the review are the comparison examples between outcomes of designs that involved older adults during the development of the tool versus those that did not, and of the results of those who involved older adults the developed tool was determined to better fit the needs of the intended population [18]. Fischer et al., explored the purposes and consequences of involving older adults in technology design, such as involvement, sense of participation, and learning. While we review similar consequences, we take this a step further and expand on the involvement processes and levels of participation. We also analyze learning as an outcome, including who benefits from the learning as well as the user testing measures employed. A previous systematic review published by Lancaster et al. focused on electronic health tools but synthesized the usability and effects [19] of the tools, rather than co-design approaches. Additionally, their target population was not defined as older adults. Another systematic review was published with similar methods and target populations as our review [1], however, they focused on assistive technology which promotes independence of living rather than on the development of EHTs. While similarities exist between our review and these, we have enough distinction from these reviews that our search will draw from a different group of studies, therefore will provide for a robust review of the current state of co-design approaches used to involve older adults in the development of EHTs.

Aim

This systematic review synthesizes the current state of utilizing co-design approaches with older adults in the development of electronic healthcare tools and seeks to answer the following questions.

- What approaches were used for involving older adults in developing EHTs and how were they defined?
- What theoretical frameworks and design principles were used to develop EHTs for older adults?
- Was iterative development utilized? If so, how many phases were conducted?
- What were the subjective and objective measures used for study endpoints, and were the measures validated?
- What involvement processes were employed?
- What is the level of participation for older adults?
- Did participants reflect on testing measures? (i.e., reflect on why an error was made)
- Was there a bi-directional flow of knowledge between participants and researchers?

Methods

Patient and Public Involvement

No patient involved

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Systematic Literature Search Strategy

We conducted a systematic review, following the Preferred Reporting Items for Systematic Reviews (PRISMA) 2020 checklist [20]. Consultations with an experienced research librarian (JB) were held to develop our search strategies. In March 2021, searches were developed for PubMed, Embase, and Scopus. A broad search strategy was utilized to ensure all co-design approaches that involve end users in the design of EHTs were captured in this review as depicted in (Supplementary Appendix A-Search Results).

The search results were screened according to the criteria defined in Table 1.

(Table 1 - Inclusion and Exclusion Criteria)

Data Extraction and Synthesis

Data extraction (selection and coding)

All records were uploaded to Covidence for screening purposes [21]. The authors (AC, KA) independently screened titles and abstracts that were identified in the search results from the database and excluded those which did not meet the selected inclusion criteria. A third reviewer (LM) resolved conflicts. After acquiring the full-text article, each author (AC, KA) independently reviewed and assessed each article to determine if it met the inclusion criteria. The final list was reviewed by the two authors to confirm which articles should remain in the review. A third reviewer (LM) resolved conflicts.

Study data were extracted for analysis and included study setting, population characteristics (age and health status), theoretical frameworks and principles used, terms used to define co-design approaches, types of electronic healthcare tools, user testing measures, types of co-design involvement processes (workshops, focus groups, interviews, prototyping, think-aloud, observation, contextual inquiry, usability testing, etc.), levels of participation, and learning outcomes.

Risk of bias (quality) assessment

While our systematic review was not focused on reported outcomes, and may not have required a quality assessment, as suggested in similar systematic reviews related to involvement processes, we wanted to provide qualifications as to how we assessed each study [18,22]. We evaluated the quality of studies included in this review, using the Agency for Healthcare Research and Quality (AHRQ) Evidence-based Practice Center approach [23]. Per this approach, two authors (AC and KA) rated studies on directness, consistency, limitations, and reporting bias, and a consensus was reached through a discussion of conflicts [23].

Strategy for data synthesis

The data from each study were combined to provide a narrative or descriptive overview. We developed a data extraction form in Excel using a random sample of 2-3 including full texts of articles and revised it iteratively. The main characteristics from the included studies are

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outlined in (Supplementary Appendix B-Study Characteristics). A comprehensive assessment of all study results has been conducted, highlighting similarities and differences in each study regarding the types of co-design approaches utilized. This comparison was necessary, as there was anticipated heterogeneity in the terminology and approaches. We used widely reported classification methods to categorize the included studies on the types of healthcare tools [17], types of involvement processes [24], levels of participation [3], and levels of learning [25,26].

To determine whether the tool described in the study met the inclusion criteria, each tool was classified using the WHO Classification of Digital Health Interventions, which is a framework that focuses on objectives within health sectors and categorizes the digital technology that fits within each sector [17]. The classification scheme focuses on interventions for 4 primary users, including clients, healthcare providers, health system or resource managers, and data services. Our target population is older adults, therefore they fit within the client framework, but it should be noted that caregivers also fit within this grouping. The WHO Classification framework was developed to synthesize evidence for digital health interventions, with the terminologies and definitions refined through public feedback, therefore is an appropriate framework for categorizing the EHTs in this review [17].

The classification tool for clients consists of 7 categories, each further broken down into subcategories, as depicted in Table 4. The tools which were developed or redesigned within each included study have been classified within 5 of the 7 overarching WHO categories, including Targeted Client Communication, Client to Client Communication, Personal Health Tracking, Citizen Based Reporting, and On-Demand Information Services to Clients.

Involvement processes were extracted from the studies and classified by the co-design framework originally developed by Leinonen and since further refined, representing 4 distinct phases, including contextual inquiry, participatory design, product design, and software prototype as hypothesis (functional prototypes) [24,27]. Levels of participation were extracted and classified based on Vaughn's framework, with levels ranging from being informed, consulted, involved, in collaboration as a co-leader, to empowering oneself and others [3]. Table 2 provides Vaughn's definitions for levels of participation. Leinonen and Vaughn's framework were combined to evaluate how the core involvement processes that span across the various co-design approaches relate to the five levels of participation, as depicted in Figure 1.

(Table 2 - Levels of Participation)

(Figure 1- Extracted from Vaughn's Involvement Framework and Leinonen's Process Framework and modified to reflect the bi-directional aspect of involvement as a process in collaboration with the stages of the co-design involvement processes.)

Within Leinonen's framework, participatory design is defined as the stage at which stakeholders have the opportunity to provide input [27]. For this literature review, this subcategory encompasses all involvement processes extracted from the studies in which

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stakeholder insight was represented. Supplementary Appendix C includes the full list of involvement processes used within each study. Other involvement processes were defined but did not fit within the cataloging of processes, which included the use of eye-tracking [8,28], teach-back methods [29], and spending time teaching older adults how to use technology in a classroom setting to better prepare them for engaging in the development process [14]. While the latter two methods do not fit within the cataloging of processes, the extent to which this involvement occurred provided a higher level of participation and associated learning outcome, suggesting that participants were empowered through these methods.

Learning outcomes were extracted from each study and classified using both Bateson's levels of learning, and Argyris' mutual learning model [25,26]. Bateson's learning framework consists of five learning levels, however only Learning Levels 0 –2 are applicable to this review [25,30,31]. Learning Level 0 refers to when a participant is provoked to respond but where no change to their action takes place. Learning Level 1 is an engaged response, in which a participant's errors are reflected on and corrected. Learning Level 2 utilizes context and tacit knowledge to draw insights for higher level learning to take place. Argyris' model consists of single and double loop learning. Single loop learning (synonymous with Learning Level 1-Reflective Learning) is the ability to detect and correct an error, which is the result of an ineffective action, and double loop (synonymous with Learning Level 2-Mutual Learning) occurs when significant actions are evaluated by the extent to which they help participants generate valid and useful information [26]. Learning Level 2-Mutual Learning is where participants, designers and researchers learn from each other, or when participants learn from reflecting on their own efforts and errors during user testing [2], including when participants are taught or exposed to technology prior to user testing [14,29]. As this review focuses on the development of EHTs, specifically the tools used to inform healthcare decisions, evaluating the role of learning through co-design approaches is necessary. If learning does not occur to the extent necessary to make qualified decisions, the information generated for consumption by the end user may not be validated.

Results

Results of Systematic Review

The initial database search returned 835 articles, including 495 from PubMed, 246 from Embase, and 94 from Scopus, of which 210 were duplicates, leaving 625 articles for the title and abstract review. Upon completion of the title and abstract review, 156 articles remained for a full-text review. The further assessment resulted in 25 studies that met the inclusion criteria. The review process is depicted in the PRISMA flow diagram [32] in Figure 2.

(Figure 2 – PRISMA Flow Diagram)

Study Quality Assessment

All included studies were found to be of high quality, with no concerns regarding directness, consistency, precision, and reporting bias. We did find some limitations, which we rated as

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"medium" level, including two studies that did not indicate frameworks [8,33], and six of the twenty-five studies had limitations due to sample size [8,11,34–37]. Overall, we did not find any studies that would rate as a "high" level limitation.

Characteristics of Included Studies

Table 3 summarizes the characteristics of all included studies. Twenty-five eligible studies were identified from our initial database search. While all studies included older adults over the age of 60, we were unable to define the overall mean age, as 6 studies simply provided the age range [7,29,37–40], and 3 simply stated participants were 65 or older [11,12,14]. Overall, the studies included participants in the age range from 60 – 91, and of the studies which provided specific ages, we have a median age of 70.9. Of the 25 studies, 13 studies involved additional participants including caregivers, healthcare providers, and others such as researchers [41,42], department heads & policymakers [42], and control group participants [43]. A total of 9 studies targeted participants with specific diagnoses (Table 3).

Of the 25 studies, the number of stakeholders across all studies ranged from 7 – 135 participants, with a mean of 35.5 for participants classified as older adults, and a mean of 40.8 when including all stakeholders.

The settings and locations varied by study. Of the 25 studies, only 11 reported a specific setting in which the study was held, including laboratories, clinics, homes, community and senior centers, as well as remote sessions via Zoom. 50% of the studies took place in North America, 46% in Europe, and only one study (4%) was from Asia.

(Table 3- Summary Characteristics)

Co-Design Approaches

Table 3 includes co-design approaches, number of phases conducted in each study and whether they defined their approach. Overall, 9 different co-design approaches were represented among the 25 studies. Of the 25 studies, 7 studies indicated the use of more than one co-design approach [8,9,37,39,41,43,44], and 2 studies stated that stakeholders were involved in development but did not indicate a specific approach [28,45]. Overall, 19 of the 25 studies conducted their study in multiple phases, ranging between 2 to 6 phases, and an overall mean of 3.4 phases per study. Of the 6 studies which did not employ iteration in development, 2 labeled user involvement as either a human factors or usability evaluation approach and did not provide definitions to the approach utilized [33,43]. Reviewing the definitions of co-design approaches, 8 studies refer to involvement with end-user/stakeholder, 5 studies refer to system usability, 34 studies refer to user needs, 3 studies refer to using prototypes, and 3 studies refer to iterative development. Studies did not determine which involvement processes were used when defining the approach.

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Theoretical Framework/Co-Design Principles

Studies did indicate that specific frameworks or principles were utilized to guide them in selecting which involvement processes to follow. Supplementary Appendix C summarizes the key theoretical frameworks or co-design principles employed for each study. Studies that reference the same terminology for both the approach and framework such as Human-Centered Design (HCD), did not use iteration in their definition, but referenced standards such as HCD Methodology – ISO 9241-210 which specifically states this is an iterative process [46].

Of the 25 studies, 2 did not indicate that their study was guided by a particular framework or principle [8,33], although one study was defined as using co-design with a participatory experience, and the other as using human factors, neither provided clarification on their approaches. As 18 different frameworks or design principles were represented across the 25 studies (Supplementary Appendix C), it is difficult to conclude which involvement processes should be included within each framework. The most commonly reported frameworks were Human Centered Design Methodology (ISO Standards) (n=5), International Patient Decision Aid Standards (IPDAS) (n=4), and User Center Design (n=3). There are inconsistencies among the studies which indicated utilizing similar frameworks or principles, such as HCD, UCD, and IPDAS, which warrants further exploration into what it means to state a study is employing a particular framework or principle. Interestingly, 6 of the 25 studies developed recommendations for frameworks that could be employed when collaboratively designing EHTs with older adults.

Electronic Healthcare Tools (EHTs)

Types of Electronic Healthcare Tools (EHTs)

Table 4 summarizes the EHTs based on the WHO Classification of Digital Health Interventions. In 12 of the 25 studies, the EHT that was developed or redesigned fits within multiple categories, therefore, was cataloged in multiple categories. Of the 25 studies, 12 developed a tool that spanned multiple categories. The overall objective was to ensure the tools included in the selected studies fit within a categorization that could be defined as those which support healthcare decisions, which was achieved through this classification process.

(Table 4 –WHO Classification of Digital Health Interventions)

Subjective and Objective Measures

We extracted 18 different measures, including 13 subjective, 3 objective, and 2 physiological measures, accounting for 19 of the 25 included studies, as indicated in Table 5. These studies utilized between 1 to 6 measures, with a mean of 3.2 measures per study, with usability as the most common endpoint, being measured in 13 of the included studies. Of the 19 studies that utilized these measures, only 9 studies used objective measures but were focused on three specific endpoints (errors, performance, and need for assistance). Within the 18 studies that used subjective measures, we extracted 23 different endpoints.

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We further categorized these measures to evaluate whether studies were using validated measures and found that only 5 studies (20% of all included studies) used validated subjective measures to evaluate their EHT. Other than the SUS scale, which was utilized in 20% of the studies to measure usability, the other validated subjective measures identified were only used in 4% - 8% of studies, while 12 distinct non-validated Likert-scale assessments spanned across 52% of all studies. Usability was also the most specified measure in non-validated Likert-scale assessments, taking place in 8 studies. Overall, 13 of the 25 studies reported measuring usability, but less than half used a validated measure.

(Table 5 – User Testing – Subjective and Objective Measures)

Stakeholder Involvement

Types of Involvement Processes

A variety of involvement processes were used in the 25 studies, as indicated in Table 6, which reports 152 processes across all the included studies. Supplementary Appendix C includes which involvement processes were used within each study. The involvement processes that were most widely reported were interviews and prototypes (medium fidelity) (n=18), followed by prototype (functional) (n=15) and the least reported was remote evaluation (n=1). Each study utilized between 2 to 11 involvement processes, with a mean of 6 processes per study.

(Table 6 – Co-Design Involvement Processes by Learning Levels)

Levels of Participation

Defining each study based on the level of participation presented challenges. Each study was assessed to have reached a participation level based on Vaughn's definitions [3] and cross-classified by co-design approach (Supplementary Appendix D). While Vaughn's definitions are related to research processes, they can be translated to the levels of participation in developing EHTs. The challenge in mapping each level against the co-design approach is that the studies did not define distinct points at which participants were involved. However, we can see that most studies employed the first three levels of participation, that of inform, consult, and involve, as evidence suggests the information provided to the research team was valued and assessed for inclusion into various design elements of the tool. However, fewer studies indicated that participants had an equal level of decision-making when making iterative changes to the design, and even fewer empowered the participants to take a lead in engaging others on the usefulness of the tool.

Figure 3 depicts each level of participation and the percent breakdown of associated involvement processes. The most distinct comparison is between the studies classified as Level 5-Empower (No) versus Empower (Yes). This comparison reflects a decrease in participatory design (from 69% to 55%) and an increase in functional prototypes (from 2% to 14%), respectively.

(Figure 3 - Involvement Processes (Percent of Total))

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Learning Outcomes

Table 3 summarizes the learning levels of each study. All 25 studies were categorized as having met learning Level 0, in which researchers benefitted from participants' responses to specific user testing measures. While 8 studies met Learning Level 1, in one of the 8 studies, only a subset of the participants were involved in reflective learning [36] as this occurred in later phases of the study, in which the number of participants had decreased by more than half. In addition to these 8 studies, one study acknowledged to the participant they had committed an error but did not provide feedback [43] and another study acknowledged it as a study limitation that they did not review results with participants [40]. 7 studies reached Learning Level 2-Mutual Learning, utilizing transference of knowledge that focused on prior experience with the technology. This benefitted both researchers and participants, indicating a bi-directional flow of knowledge occurred. It should be noted that these 7 studies were also classified as having reached the empower participation level.

Table 6 summarizes the difference in the types of involvement processes used when studies engaged researchers and participants in mutual learning. While additional studies engaged in observational activities that provided opportunities for additional context, these studies did not engage in mutual learning. For example, an additional 2 studies utilized contextual inquiry, however, in these studies the learning was unidirectional and only benefitted the researchers.

Discussion

Key Findings

This review focused on synthesizing the current state of involving older adults in co-design approaches to develop EHTs and aimed to identify how the co-design approaches utilized mutual learning techniques to benefit older adult participants. We extracted evidence from each study to categorize the terms used to describe co-design approaches, as well as to draw themes around how these terms were defined. We identified the EHTs and categorized them by health sector classifications as well as the subjective and objective measures employed. This review also explored the frameworks and principles that determined the stakeholder involvement, including types of involvement processes, levels of participation and learning outcomes.

Co-Design Approaches

As anticipated, heterogeneity exists within the definitions used to describe co-design approaches. While 13 definitions were represented across the studies, general themes presented suggest that the approaches could be used interchangeably. While this may further exacerbate a problem, and lead to inconsistent use of terminology, what is of notable importance within this review, is not so much on the name of the approaches used, but how they are interpreted and then employed to determine the methods for involving and co-designing with older adults. Not surprisingly, each study specified either definitively within their

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aim or subsequently, that end-users should have some level of involvement or participation during the development life cycle process. However, studies did not specify the extent to which older adults should be involved, nor did they specify whether a learning model should be employed.

Frameworks and Principles

We chose not to cross-classify this variable as to not misrepresent relationships; however, the diversity of frameworks extracted exemplifies a need for a framework that is specific to engaging older adults in the development of EHTs, which can be consistently utilized, and which also provides validation to the involvement processes that should be employed. This need is further exemplified as 25% (n=6) of the included studies suggested recommendations for a framework that could be used when engaging older adults in the development of EHTs. Further research is needed to determine whether the proposed, or a combination of existing and proposed frameworks can be validated to optimize older adults' involvement, one that provides an environment for mutual learning and empowerment.

Electronic Healthcare Tools (EHTs)

While the EHTs that were applied in the studies spanned multiple categories under the WHO classification, the majority focused on tools that provided the ability to search for information on a particular health topic or that allowed participants to track their health information through self-monitoring, including accessing personal health records, use of wearables, or documenting health information. Each study involved participants in developing an EHT that would support older adults' healthcare decisions. Previous studies have reported positive impacts to both older adults and improvements in technology when engaging them in the development lifecycle [1,2].

Furthering the notion that a framework, specific to older adults in developing EHTs, needs to be evaluated, is concerning the inconsistency of user testing assessments. We extracted evidence that 19 different measures were utilized in 76% (n=19) of the included studies and found that only five studies (20% of all included studies) used a validated subjective measure to evaluate their EHT. Usability was the most specified measure, and while the SUS, a validated usability assessment was utilized, more than half of the studies utilized a non-validated Likert-scale assessment. This discrepancy in assessments for user testing makes it difficult to draw definitive conclusions as to whether studies are consistently measuring what they intended to measure. This is consistent with a recent review that found studies often use non-validated questionnaires rather than readily available validated assessments and that despite recommendations to measure learning, this is rarely employed [47].

Stakeholder Involvement, Participation and Mutual Learning

While we found inconsistency in the involvement processes among the studies, over 70% included interviews and prototyping, others included a range of processes such as cognitive walkthroughs, think-aloud sessions, surveys, scenarios, and focus groups, and only 20% of the

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studies utilized contextual inquiries and use cases. While fewer studies utilized contextual inquiries, the studies that employed this process were more likely to be categorized as having both empowered users and involved them in mutual learning. For studies in which participants and researchers benefitted from mutual learning, we observed a higher utilization of specific involvement processes including contextual inquiry, cognitive walkthroughs, focus groups, working group workshops and functional prototyping, and a decrease in think aloud sessions. This is consistent with previous research by both Bateson and Argyris as well as recent studies that evaluated learning outcomes [30,48] and suggests the importance of understanding user context and involvement in group activities that facilitate reflective and mutual learning. While Argyris also suggests that when groups work on important topics, they are less receptive to corrective feedback, inhibiting learning and potentially producing information of lower value; we observed that these processes were effectively employed and solicited valid information on important topics, or that of improving the development of EHTs that are meant for healthcare decisions.

To the best of our knowledge, current frameworks, or principles have not provided guidance on distinguishing the levels of participation, nor the validated involvement processes that should be utilized when engaging older adults in developing EHTs. To provide structure to this notion, this review drew upon Leinonen's framework for processes and Vaughn's framework for levels of participation. After extracting evidence of both variables from the 25 studies, and overlaying these on top of the two frameworks, we began to recognize the areas in which we need to further explore. We need to explore what it means to involve older adults, specifically which involvement processes are more likely to lead to empowerment and mutual learning. This is important because the evidence extracted from the studies suggests that bi-directional knowledge, which benefits both older adults and researchers, may increase as participation levels increase.

It has also been suggested that exposure to the technology before usability testing has shown to be an effective means of eliciting users' needs, such that the increased knowledge and awareness of the tool inspires ideation and innovation [9,37], and that tacit knowledge or real-life context increases the ability to portray specific user needs [42]. However, there are concerns about bias if an individual is exposed to technology before soliciting their design ideas [49]. Not to dismiss the concern for bias, but further evidence supports the notion that experienced users of technology can provide more meaningful insight during design sessions and could provide feedback more representative of the target population [50]. We also see that allowing users to engage with the technology for an extended time or ensuring they understand how to use the technology, reduces frustration when contributing to design ideas [14,29,49]. For example, when older adults were asked to participate in two similar design activities, they were more involved when they had familiarity with the topic, compared to being asked to innovate on a novel topic [34].

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Suggestions have been made to separate co-design processes from outcomes and focus on the role that participants have to be co-designers and co-initiators [51], which supports the values defined by the fourth and fifth levels of involvement, that of collaborate and empower, respectively. We should be looking at involvement as a process, one in which through various stages of the development life cycle, users have the potential to cross through the spectrum of involvement highlighted in Vaughn's framework. By combining Leinonen and Vaughn's framework, we can see how the involvement processes that span across the various co-design approaches, can be used in iterative development, but through these various stages, users can participate in the processes at different levels, from being informed, consulted, involved, in collaboration or as a co-leader, to empowering oneself and others. This is consistent with a review by Fischer et al. that also suggest that involvement as a process should be a consideration for future research and policy, specifically the extent to which users are involved and receive benefit [18]. Further, we suggest that additional exploration into the extent to which a greater understanding of the tool can contribute to richer feedback is necessary [9]. We also suggest the levels of participation must be bi-directional for mutual learning to occur, as depicted in Figure 1, and when realized within the development lifecycle will result in greater contributions to the development of the tool.

Limitations

As our inclusion criteria spanned an array of terms that we categorized as co-design approaches, we had several approaches that were only represented by one study, which may not fully represent all studies that incorporate such approaches. One of the biggest limitations of this study was the difficulty in extracting and classifying the information, however we used mitigation strategies (e.g., established frameworks and classification methods) to support the review of variables including types of EHTs, stakeholder involvement and participation and levels of learning.

Conclusions

This review depicted not only the heterogeneity in the terminology used to define co-design approaches, but also the gaps in frameworks and design principles for developing EHTs for older adults. Most studies in this review developed EHTs specific to patient engagement and designed their study to include older adults' feedback in the design, indicating a collaborative focus centered on meeting the needs of this population. However, we found that studies did not consistently utilize validated assessments for conducting usability testing. We also found gaps in the evidence to support which involvement processes should be used when co-designing with older adults. Additionally, gaps exist around which learning models should be employed when conducting user testing with older adults. Future research needs to evaluate how empowerment is associated with mutual learning, whether this bi-directional knowledge can occur at all participation levels, and whether the benefit to both older adults and researchers increases as the levels of participation increases. To ensure studies are optimizing the involvement processes, participation and learning of older adults when developing EHTs,

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improving existing co-design frameworks to meet the needs of older adults should be evaluated.

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Contributors

AC and KA conceived of the broad concept for the literature review. AC developed the protocol, then worked with KA and JB to develop a search strategy. AC and KA independently screened titles and abstracts that were identified in the search and excluded those which did not meet the selected inclusion criteria. LM resolved conflicts. After acquiring the full-text articles, AC and KA independently reviewed and assessed each article to determine if it met the inclusion criteria. The final list was reviewed by the AC and KA to confirm which articles should remain in the review. A third reviewer LM resolved conflicts. AC extracted and synthesized the data. AC wrote the review, with meaningful insight in the structure of the analysis and conclusion with KA and LM. Final review of the manuscript by KA, LM, AK, and DR.

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Competing Interests

None declared

Patient consent for publication

Not required

Ethics Statement

This is a systematic review, without human involvement, as such an ethical assessment was not required.

Data availability statement

All data relevant to the study are included in the article or uploaded as supplemental information.

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Figures

Figure 1- Extracted from Vaughn's Involvement Framework and Leinonen's Process Framework and modified to reflect the bi-directional aspect of involvement as a process in collaboration with the stages of the co-design involvement processes.

Figure 2 – PRISMA Flow Diagram

Figure 3 - Involvement Processes (Percent of Total)

Tables

Table 1 - Inclusion and Exclusion Criteria

Selection Criteria	*Inclusion Criteria*	*Exclusion Criteria*
Participants	Older adults aged 60 or older, without regard to gender, race, or ethnicity	If the study population included adults of all ages, including greater than 60 years of age, but evidence could not be extracted from those greater than 60 years of age, the study was excluded
Study Setting	All study settings were included	
Study Design	All study designs were included	
Methods and Tools	Studies published that employed co-design approaches in the development of electronic healthcare tools	Studies that used non-electronic healthcare tools
Language		Studies in which publications are not in the English language were excluded

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Table 2 - Levels of Participation

Levels of Participation	***Definitions-Vaughn et al. [3]***
Inform	Information is shared with community members and could become more participatory if members ask for information relative to their interests.
Consult	Participants provide feedback to the researchers, as related to a specific decision point.
Involve	Participants provide feedback to researchers during the entire process.
Collaborate	Participant input is valued equally, as a co-leader in the development process.
Empower	Participants take a key role in leading the initiative to get others engaged.

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Table 3- Summary Characteristics

First Author's last name	Year of Pub	First author's location	Study setting	Co-Design Approach	Co-Design Approach Defined	No. of Phases	Learning Level	Learning Beneficiary	No. of Stakeholders	No. of OA	OA/P	Ca	HP	Ot	Age of older adults (mean)	Diagnosis
Abujarad [52]	2021	USA	Yale School of Medicine	User Centered Design	Yes	-	Level 0	Research Team	38	24	X	X	X		69	
Ahmed [34]	2019	USA		Participatory Design	Yes	-	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	9	7	X	X			67	Heart Failure
Bogza [29]	2020	Canada	Clinics, research center, or in their own homes.	User Centered Design	No	6	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	19	12	X	X	X	X	(2) 60 – 64 years (6) 65 – 74 years, (3) 75 – 84 years, (1) >85 years	Mild Cognitive Impairment (MCI)
Cornet [12]	2020	USA	Laboratory	User Centered Design	Yes	3	Level 0	Research Team	63	50	X	X	X	X	>65	Heart Failure
Gonzalez [33]	2014	USA		Human Factors	No	-	Level 0	Research Team	12	12	X				77.7	
Grimaldi [11]	2020	Italy		Human Centered Design	Yes	4	Level 0	Research Team	10	10	X				> 60	
Gustafson [14]	2016	USA	Home visits	User Centered Design	Yes	5	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	135	135	X	X	X	X	> 65	
Harrington [9]	2018	USA		Co-Design Human Centered Computing Interaction Design Process Participatory design	No	2	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	39	39	X				72.1	
Harte [7]	2017	Ireland	Participant's home	Human Centered Design	Yes	3	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	12	12	X				61-85	

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First Author's last name	Year of Pub	First author's location	Study setting	Co-Design Approach	Co-Design Approach Defined	No. of Phases	Learning Level	Learning Beneficiary	No. of Stakeholders	No. of OA	OA/P	Ca	HP	Ot	Age of older adults (mean)	Diagnosis
Hoffman [39]	2020	USA		Co-Design User Centered Design	No	3	Level 0	Research Team	16	8	X	X			Phase 2: 65-84 Phase 3: 61-89	
Holden [53]	2020	USA	Eskenazi Health	User Centered Design	Yes	3	Level 0	Research Team	23	23	X				67.6	
Kim [38]	2018	USA	Participant's home or senior center	Participatory Design	No	2	Level 0	Research Team	17	17	X				65-80	
Mansson [35]	2020	Sweden		Co-creation	Yes	2	Level 1 Sgl Loop	Research Team Participant	10		X				76	
Martin-Hammond [36]	2015	USA	On-site 2 Locations	User Centered Design	Yes	3	Level 0	Research Team	7	7	X				70	
Nguyen [42]	2019	Netherlands		Co-Design	No	3	Level 0	Research Team	56	56	X			X	68.9	
Nielsen [44]	2018	Denmark	Eriksholm Research Centre	Participatory Design Co-Design	Yes	3	Level 0	Research Team	54	36	X	X	X		70.3	Hearing impairment
Or [43]	2012	Hong Kong	Community health service center	Usability Evaluation Approach Usability Test Design	No	-	Level 0	Research Team	57	50	X			X	71.6	One of the following: hypertension, diabetes, heart disease, asthma, prostatitis, hypotension
Petersen [45]	2019	Netherlands		Does Not State	No	3	Level 0	Research Team	71	32	X		X		70	Larynx cancer
Poirier [28]	2019	USA		Does Not State	No	-	Level 0	Research Team	45	45	X				72.2	
Portz [40]	2020	USA	Home, clinic, or zoom	Human Centered Design	Yes	3	Level 0	Research Team	81	31	X	X	X		66-91	Heart Failure (palliative care patients)
Scandurra [37]	2013	Sweden		Participatory Design User Centered Design	No	5	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	8	8	X				65-80	

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First Author's last name	Year of Pub	First author's location	Study setting	Co-Design Approach	Co-Design Approach Defined	No. of Phases	Learning Level	Learning Beneficiary	No. of Stakeholders	No. of OA	OA/P	Ca	HP	Ot	Age of older adults (mean)	Diagnosis
Ummels [41]	2020	Netherlands		Co-creation User Centered design	Yes	4	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	63	63	X			X	70.3	
Van De Dijk [54]	2013	Netherlands		User Centered Design	Yes	-	Level 0	Research Team	86	86	X				72.5	COPD
Van Velsen [55]	2019	Netherlands		Participatory Design	No	2	Level 0	Research Team	82	82	X	X			71.5	
Wikberg-Nilsson [8]	2018	Sweden		Co-Design Participatory User Experience Design	No	5	Level 1 Sgl Loop	Research Team Participant	7	7	X				75	Reduced hearing, eyesight, mobility, sensibility, and loss of memory function

*OA/P = Older Adults/Patients, Ca = Caregivers, HP = Healthcare Providers, Ot = Others

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Table 4 –WHO Classification of Digital Health Interventions

WHO Classification of Digital Health Interventions	*Total studies (n=25), n(%)*	*References*
1.1 Targeted Client Communication		
1.1.2 Transmit targeted health information to client(s)	1 (4%)	[44]
1.1.3 Transmit targeted alerts and reminders	2 (8%)	[37,43]
1.1.4 Transmit diagnostics result, or availability	1 (4%)	[34]
1.3 Client to Client Communication		
1.3.1 Peer Group for Clients	3 (12%)	[14,37,44]
1.4 Personal Health Tracking		
1.4.1 Access by client to own medical records	3 (12%)	[34,35,38]
1.4.2 Self-monitoring of health or diagnostic data by client	7 (28%)	[7,33,34,41,44,52,54]
1.4.3 Active data capture/documentation by client	8 (32%)	[9,12,29,34,35,39,40,43]
1.5 Citizen Based Reporting		
1.5.1 Reporting of health system feedback by clients	2 (8%)	[52,55]
1.6 On-Demand Information Services to Clients		
1.6.1 Client Look-up of health information	16 (64%)	[8,11,12,28,29,36–40,42,44,45,52,53,55]

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Table 5 – User Testing – Subjective and Objective Measures

Measures	Endpoint	Total studies (n=25) n (%)	References
Validated			
Subjective Measures			
ACE (Altarum Consumer Engagement)	Patient engagement	1 (4%)	[34]
ASQ	Efficiency	1 (4%)	[7]
CES	Confidence & Computer Efficacy	1 (4%)	[52]
Decisional Conflict Scale (values clarity)	Values Clarity	2 (8%)	[28,29]
I-PANAS-SF	Emotional Reactions	1 (4%)	[52]
Likert scale (specified validity)	Comprehension/Recall, Graphicacy, Numeracy	1 (4%)	[28]
NASA-TLX	Cognitive Workload	2 (8%)	[7,12]
NVS	Health Literacy	2 (8%)	[12,34]
Ottawa Acceptability Scale	Quality	2 (8%)	[29,39]
SURE scale	Values Clarity	1 (4%)	[39]
SUS	Usability	5 (20%)	[7,12,29,52,53]
Us.E 2.0	Quality	1 (4%)	[11]
Non-Validated			
Subjective Measures			
Likert scale (non-specified validity)[a]		13 (52%)	[7,9,11,12,14,29,33,37,41–45]
Objective Measures			
Mistakes Counted	Errors	7 (28%)	[7,11,12,39,42,43,53]
Time to Complete	Performance	4 (12%)	[7,28,43,53]
Times Request Assistance	Need for Assistance	2 (8%)	[43,53]
Physiological Measures (eye-tracking)			
Cognitive Processes	Usability	1 (4%)	[8,28]
Gaze Fixation	Usability	1 (4%)	[8,28]

Satisfaction, User Acceptance, Appearance, Comfort, Functionality, Reliability, Usability, Usefulness, Ease of Use, Comprehensibility, Effectiveness, Value of Information [a]

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Table 6 – Co-Design Involvement Processes by Learning Levels

■ Contextual Inquiry ■ Participatory Design ■ Product Design ■ Prototype ■ Other

(color-coded to reflect association in Figure 3)

	Involvement Processes	*Level 0 (n=25), n(%)*	*Single Loop Level 1 (n=8), n(%)*	*Double Loop Level 2 (n=7), n(%)*
Contextual Inquiry	Contextual Inquiry	6 (24%)	4 (50%)	4 (57%)
Participatory Design	Survey	10 (40%)	3 (38%)	3 (43%)
	Interview	18 (72%)	7 (88%)	5 (71%)
	Cognitive walkthrough	8 (32%)	4 (50%)	3 (43%)
	Think aloud	11 (44%)	2 (25%)	1 (14%)
	Remote evaluation	1 (4%)	1 (13%)	1 (14%)
	Focus group	8 (32%)	4 (50%)	4 (57%)
	Task analysis	10 (40%)	3 (38%)	2 (29%)
	Scenarios	7 (28%)	2 (25%)	1 (14%)
	Prototype (low fidelity)	8 (32%)	4 (50%)	3 (43%)
	Working Group Workshop	9 (36%)	5 (63%)	3 (43%)
Product Design	Use Cases	4 (16%)	2 (25%)	1 (14%)
	User Stories	9 (36%)	2 (25%)	2 (29%)
	Prototype (medium fidelity/throw-away)	18 (72%)	8 (100%)	5 (71%)
	Heuristic evaluation	5 (20%)	2 (25%)	1 (14%)
Software Prototype as Hypothesis	Prototype (functional/Beta version)	15 (60%)	8 (100%)	7 (100%)
Other	Other	5 (16%)	3 (38%)	2 (29%)
Total		**152**	**64**	**48**

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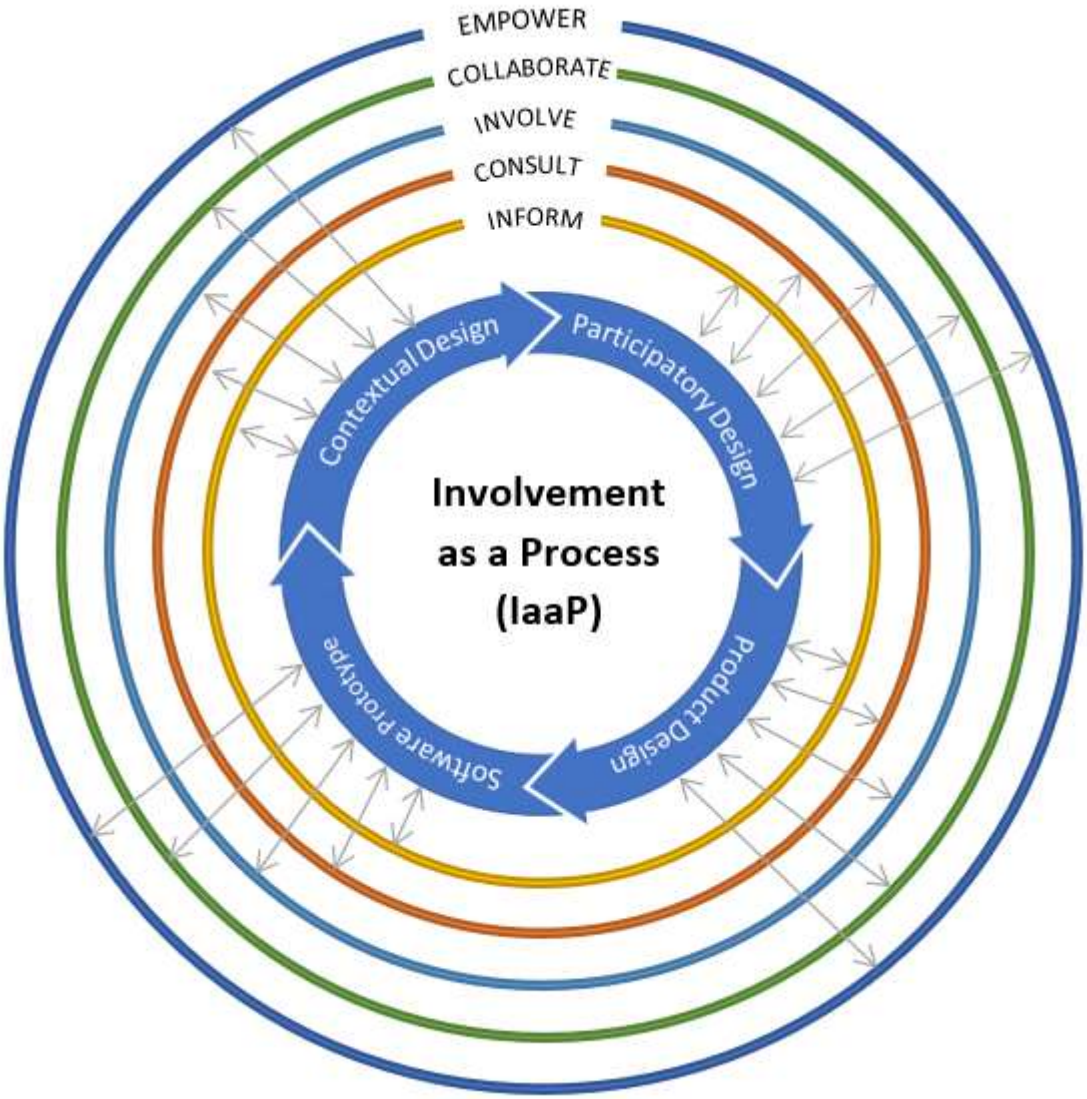


Figure 1- Extracted from Vaughn's Involvement Framework and Leinonen's Process Framework and modified to reflect the bi-directional aspect of involvement as a process in collaboration with the stages of the co-design involvement processes.

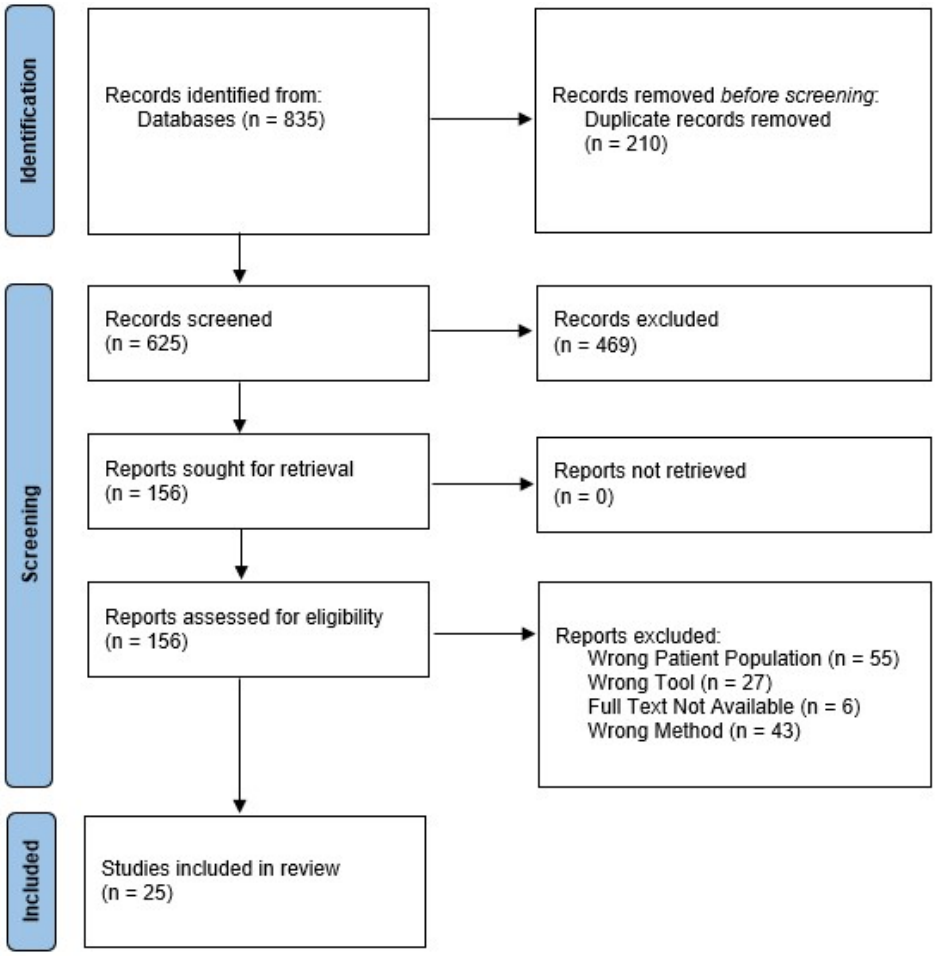


Figure 2 – PRISMA Flow Diagram

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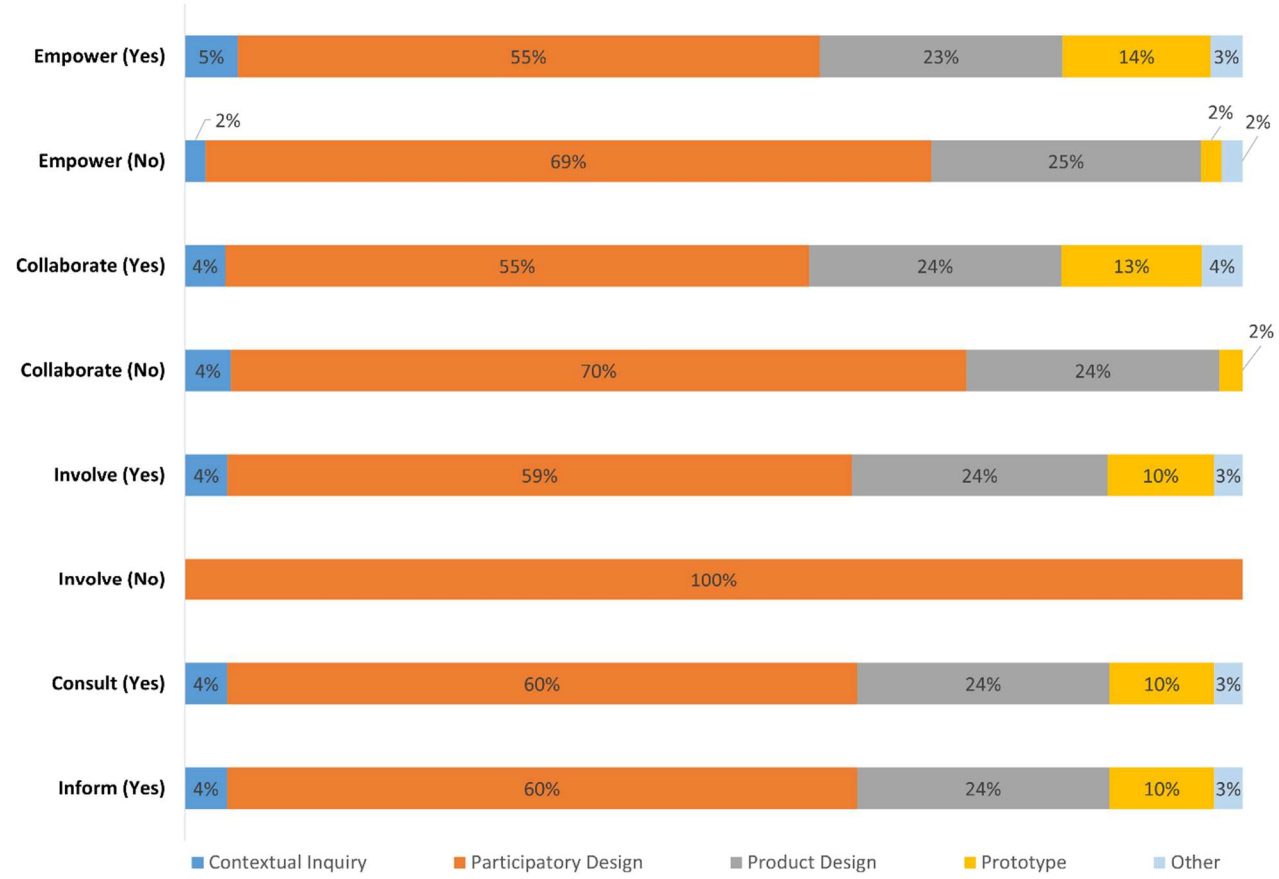


Figure 3 - Involvement Processes (Percent of Total)

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1 Supplementary Appendix A - Search Results

PubMed – 3.17.2021		
Search #	Query	Results
#1	(Co-Design[tiab] OR Co-Creation[tiab] OR Co-Production[tiab] OR Participatory Design[tiab] OR Human Factors[tiab] OR User-Centered Design[mesh] OR User-Centered Design[tiab] OR Participatory Approach[tiab] OR Human-Centered Design[tiab] OR Universal design[mesh] OR Patient Participation[mesh] OR patient participation[tiab])	38,538
#2	(Aged[mesh] OR Older Adults[tiab] OR Older People[tiab] OR Elderly[tiab] OR Geriatric[tiab] OR Aged over 60[tiab] OR aged 60[tiab] OR Seniors[tiab] OR Over 65[tiab])	3,333,143
#3	(Electronic Health Records[mesh] OR Electronic Health Records[tiab] OR Electronic Health Record[tiab] OR Electronic Healthcare[tiab] OR eHealth[tiab] OR "mHealth"[tiab] OR Decision Aid[tiab] OR decision aids[tiab] OR Electronic Healthcare Aid[tiab] OR Healthcare Application[tiab] OR digital health[tiab] OR health information technology[tiab] OR health information technologies[tiab] OR health technology[tiab] OR health technologies[tiab])	59,271
#4	#1 AND #2 AND #3	495

Embase – 3.17.2021		
Search #	Query	Results
#1	('human factors research'/exp OR 'user-centered design'/exp OR 'Universal design'/exp OR 'Patient Participation'/exp OR ('co-design' OR 'Co-Creation' OR 'Co-Production' OR 'Participatory Design' OR 'Human Factors' OR 'user-centered design' OR 'Participatory Approach' OR 'Human-Centered Design' OR 'patient participation'):ab,ti)	42,464
#2	('aged'/exp OR ('older adult' OR 'older adults' OR 'older people' OR elderly OR geriatric OR geriatrics OR 'aged over 60' OR 'aged 60' OR 'over 60' OR 'aged over 65' OR 'aged 65' OR 'over 65' OR senior OR seniors):ab,ti)	3,460,578
#3	('electronic health record'/exp OR ('electronic health record' OR 'electronic health records' OR 'electronic healthcare' OR 'ehealth' OR 'mhealth' OR 'electronic decision aid' OR 'electronic decision aids' OR 'electronic healthcare aid' OR 'electronic healthcare aids' OR 'healthcare applications' OR 'Decision Aid' OR 'decision aids' OR 'digital health' OR 'health information technology' OR 'health information technologies' OR 'health technology' OR 'health technologies'):ab,ti)	59,924
#4	#1 AND #2 AND #3	246

Scopus – 3.17.2021		
Search #	Query	Results
#1	TITLE-ABS("co-design" OR "Co-Creation" OR "Co-Production" OR "Participatory Design" OR "Human Factors" OR "user-centered design" OR "Participatory Approach" OR "Human-Centered Design" OR "patient participation")	78,052
#2	TITLE-ABS("older adult" OR "older adults" OR "older people" OR elderly OR geriatric OR geriatrics OR "aged over 60" OR "aged 60" OR "over 60" OR "aged over 65" OR "aged 65" OR "over 65" OR senior OR seniors)	657,056

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#3	TITLE-ABS("electronic health record" OR "electronic health records" OR "electronic healthcare" OR "ehealth" OR "mhealth" OR "electronic decision aid" OR "electronic decision aids" OR "electronic healthcare aid" OR "electronic healthcare aids" OR "healthcare applications" OR "Decision Aid" OR "decision aids" OR "digital health" OR "health information technology" OR "health information technologies" OR "health technology" OR "health technologies")	54,356
#4	#1 AND #2 AND #3	94

- records before duplicates removed: 835
- records after duplicates removed: 625

2 Supplementary Appendix B - Study Characteristics

Category	*Variable*	*Classification method*
Summary characteristics	Year of publication	Metadata
	First author's geolocation	Metadata
	Study setting	Metadata
	Number of stakeholders	Metadata, mean
	Number of older adults	Metadata, mean
	Study population	Metadata
Co-design approaches	Terminology or definitions	Manual, binary
	Theoretical framework/design principles	Manual, binary
	Iterative development	Metadata, mean
Stakeholder involvement	Participant types	Metadata
	Types of Involvement Processes	Leinonen et.al [24]
	Learning Levels	Bateson [25], Argyris [26]
	Levels of participation	Vaughn et.al [3]
Electronic healthcare tools	Types of healthcare tools	WHO classification of digital health interventions [17]
	User testing	Metadata

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3 Supplementary Appendix C

Author	Theoretical Framework/Co-Design Principle	Involvement Processes																	User Testing Measures and Validation					
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	Endpoint	Other	Objective Measure	Subjective Measure	Physiological	Validated
Abujarad	User Centered Design Approach	X	X	X	X	X			X					X	X		X		confidence and computer efficacy			CES		X
																			emotional reactions			I-PANAS-SF		X
																			usability			SUS		X
Ahmed	Participatory Design Approach Developed recommendations for framework	X	X					X			X	X			X		X		health literacy			NVS		X
																			patient engagement - self report on health behaviors			ACE (Altarum Consumer Engagement)		X
Bogza	Center for eHealth and Wellbeing Research guidelines Inductive thematic qualitative analysis approach IPDAS			X	X	X								X	X		X	X	quality of decision aid			Ottawa Acceptability Scale		X
																			satisfaction			Likart - unspecified		
																			usability			SUS		X
																						Decisional Conflict Scale		X
																			Comprehension	teach back method				
Cornet	User Centered Design Developed recommendations for framework		X	X	X	X			X	X			X		X	X			cognitive workload			NASA-TLX		X
																			health literacy			NVS		X
																			usability			SUS		X
																			user acceptance			Likart - unspecified		
																			errors		Not stated			
Gonzalez	none																		appearance			Likart - unspecified		
																			comfort			Likart - unspecified		
				X											X				functionality			Likart - unspecified		
																			reliability			Likart - unspecified		
																			usability			Likart - unspecified		
Grimaldi	Human computer design approach ISO 13407:1999 [1] revised by ISO 9241-210:2010		X	X	X	X			X				X		X		X		quality of interaction			Us.E 2.0		
																			satisfaction			Likart - unspecified		
																			usability			Likart - unspecified		
																			errors		Not stated			
Gustafson	NIATx model as a framework Developed recommendations for framework	X		X	X			X			X			X	X		X	X	usability	Spending time teaching older adults how to use technology in a classroom setting		Likart - unspecified		

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Author	Theoretical Framework/Co-Design Principle	Involvement Processes																	User Testing Measures and Validation					
Harrington	Experience-based co-design Developed recommendations for framework	X	X	X				X				X					X		usability			Likart - unspecified		
																			usefulness	Likart - unspecified				
Harte	HCD methodology ISO 9241-210 standards	X	X	X	X				X	X	X		X		X	X	X		cognitive workload			NASA-TLX		X
																			efficiency			ASQ		X
																			satisfaction			Likart - unspecified		
																			usability			SUS		X
																			errors		Mistakes Counted			
																			Performance		Time to Complete			
Hoffman	IPDAS Ottawa Decision Support		X	X	X	X		X		X	X			X	X	X	X		quality of decision aid			Ottawa Acceptability Scale		X
																			values clarity			SURE		X
																			errors		Observed errors			
Holden	Age-Related Research-Based Usability Guidelines Principles and Creative Human Factors Approaches		X	X		X			X					X	X		X		usability			SUS		X
																			errors		Mistakes Counted			
																			need for assistance		Times Request Assistance			
																			Performance		Time to Complete			
Kim	Grounded theory approach	X	X	X				X			X	X		X										
Mansson	Karagianni's Optimized Honeycomb model					X			X			X			X		X		findable, accessible, usable, desirable, credible, useful, and valuable	UX Honeycomb Model ALL QUALITATIVE				
Martin-Hammond	Nielsen's Heuristics			X	X					X	X		X		X	X			usability	qualitative				
Nguyen	Medical Research Council framework Developed recommendations for framework			X		X			X			X		X	X		X		satisfaction			Likart - unspecified		
																			errors		Observed errors			
																			Navigation	Google Analytics				
Nielsen	HCD International Organization for Standardization (ISO) standards for human-centered design for interactive systems (European Committee for Standardization, 2010) and applied principles and techniques from the user innovation management method (Kanstrup & Bertelsen, 2011).							X			X			X	X		X		usability			Likart - unspecified		
																			user acceptance			Likart - unspecified		
Or	Human Factors Design principles			X		X			X						X	X			ease of use			Likart -		

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Author	Theoretical Framework/Co-Design Principle	Involvement Processes																	User Testing Measures and Validation					
																						unspecified		
	Nielson heuristic principles Systematic usability study guideline																		satisfaction			Likart - unspecified		
																			usefulness			Likart - unspecified		
																			errors		Mistakes Counted			
																			Need for assistance		Times Request Assistance			
																			Performance		Time to Complete			
Petersen	IPDAS																		comprehensibility			Likart - unspecified		
																			effectiveness			Likart - unspecified		
				X		X								X		X			satisfaction			Likart - unspecified		
																			usability			Likart - unspecified		
																			value of the information			Likart - unspecified		
Poirier	IPDAS																		comprehension/recall			Likart - specified validity		X
																			gaze fixation				eye-tracking	
																			graphicacy			Likart - specified validity		X
									X	X							X		numeracy			Likart - specified validity		X
																			values clarity			Decisional Conflict Scale		X
																			performance		Time to Complete			
Portz	Grounded theory approach Human-Centered Design framework			X		X																		
Scandurra	Human-Computer Interaction (HCI) research theory											X							share experience with others			Likart - unspecified		
																	X		usability			Likart - unspecified		
																			user acceptance			Likart - unspecified		
Ummels	Double Diamond Method MoSCoW model Ummels Framework Developed recommendations for framework			X			X	X	X					X		X			satisfaction			Likart - unspecified		
																			usability			Likart - unspecified		
Van De Dijk	User Centered Design Approach		X					X		X		X						X						

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Author	Theoretical Framework/Co-Design Principle	Involvement Processes																User Testing Measures and Validation					
Van Velsen	MoSCoW method								X	X	X		X					usability	qualitative				
Wikberg-Nilsson	None			X							X			X			X	cognitive process				eye-tracking	
																		usability	qualitative				

A. Contextual Inquiry
B. Survey
C. Interview
D. Cognitive walkthrough
E. Think aloud
F. Remote evaluation
G. Focus group
H. Task analysis
I. Scenarios
J. Prototype (low fidelity)
K. Working Group Workshop
L. Use Cases
M. User Stories
N. Prototype (medium fidelity)
O. Heuristic evaluation
P. Prototype (high fidelity – functional/Beta version)
Q. Other Process

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4 Supplementary Appendix D

Co-Design Approach	***Inform***		***Consult***		***Involve***		***Collaborate***		***Empower***	
	n	ref	n	ref	n	ref	n	ref	n	ref
Co-creation (n=2)	**2**	[35,41]	**2**	[35,41]	**2**	[35,41]	**2**	[35,41]	**2**	[35,41]
Co-design (n=5)	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]
Human centered computing (n=1)	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]
Human Centered Design (HCD) (n=3)	**3**	[7,11,40]	**3**	[7,11,40]	**3**	[7,11,40]	**2**	[7,11]	**1**	[7]
Human Factors (n=1)	**1**	[33]	**1**	[33]	**1**	[33]	**0**		**0**	
Interaction Design Process (n=1)	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]
Participatory Design (n=7)	**7**	[8,9,34,37,38,44,55]	**7**	[8,9,34,37,38,44,55]	**7**	[8,9,34,37,38,44,55]	**5**	[8,9,34,37,44]	**5**	[8,9,34,37,44]
Usability Evaluation Approach (n=1)	**1**	[43]	**1**	[43]	**1**	[43]	**1**	[43]	**0**	
User Centered Design/Approach (n=9)	**9**	[12,14,29,36,39,41,52–54]	**9**	[12,14,29,36,39,41,52–54]	**9**	[12,14,29,36,39,41,52–54]	**5**	[14,29,39,41,53]	**6**	[14,29,39,41,52,53]
Other (Did not Specify) (n=2)	**2**	[28,45]	**2**	[28,45]	**1**	[45]	**1**	[45]	**1**	[45]

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2-4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	4-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	5-6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	4-6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	5
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	NA
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	4-6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	4-6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	5
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	NA
Study characteristics	17	Cite each included study and present its characteristics.	6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	8
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	6-11
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	NA
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	11-13
	23b	Discuss any limitations of the evidence included in the review.	14
	23c	Discuss any limitations of the review processes used.	14
	23d	Discuss implications of the results for practice, policy, and future research.	14
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	15
Competing interests	26	Declare any competing interests of review authors.	15
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71



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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	No
Registration	12	Provide the register name and registration number.	Yes

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Co-design approaches involving older adults in the development of electronic healthcare tools: a systematic review

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Co-design approaches involving older adults in the development of electronic healthcare tools: a systematic review

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Word Count: 5,777

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Co-design approaches involving older adults in the development of electronic healthcare tools: a systematic review

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Abstract

Objective

The primary aim was to review and synthesize the current evidence of how older adults are involved in co-design approaches to develop electronic healthcare tools (EHTs). The secondary aim was to identify how the co-design approaches utilized mutual learning techniques to benefit older adult participants.

Design

Systematic review following the Preferred Reporting Items for Systematic Reviews (PRISMA) 2020 checklist.

Data Sources

PubMed, Embase, and Scopus databases were searched for studies from January 2010 to March 2021.

Eligibility Criteria

Inclusion criteria were studies employing co-design approaches to develop an EHTs, and the study population was aged 60 years and older.

Data Extraction and Synthesis

Data were extracted for analysis and risk of bias. We evaluated the quality of studies using the Agency for Healthcare Research and Quality (AHRQ) Evidence-based Practice Center approach.

Results

Twenty-five studies met the inclusion criteria for this review. All studies utilized at least 2 involvement processes, with interviews and prototypes used most frequently. Through cross-classification, we found an increased utilization of functional prototypes in studies reaching the "empower" level of participation and found that studies which benefitted from mutual learning

had a higher utilization of specific involvement processes such as focus groups and functional prototyping.

Conclusions

We found gaps to support which involvement processes, participation levels and learning models should be employed when co-designing with older adults. This is important because higher levels of participation may increase the user's knowledge of technology, enhance learning, and empower participants. To ensure studies optimize participation and learning of older adults when developing EHTs, there is a need to place more emphasis on the approaches promoting mutual learning.

Systematic Review Registration

PROSPERO CRD42021240013

Strengths and limitations of this study

- This review provided a comprehensive assessment of the utilization of co-design approaches in studies that involve older adults in developing EHTs.
- We developed a broad search strategy to ensure all co-design or "collaborative" approaches were captured in this review.
- We utilized widely reported classification methods to categorize the included studies.
- We limited our analysis to the overall involvement processes and levels of participation at the study level; cross-classification of these variables at the design stage level for each study was not conducted.
- This review was limited to initial studies and did not evaluate whether subsequent papers by research groups provided a broader analysis of user experience.

Introduction

Co-design approaches aim to elicit ideas and foster a non-hierarchical environment for stakeholders, in which their concepts, tacit knowledge, and lived experiences can be applied to develop tools that meet their needs, improve usability and impact outcomes from use of the system [1,2]. Co-design is also likened to participatory design, each defined as focusing on the engagement and creativity of stakeholders, or those untrained in design efforts, but through collaborative efforts bring mutual learning into the development process [3,4]. Despite the underlying similarity of these approaches, there is heterogeneity in the terminology of co-design approaches, which leads to variation in the evaluation measures employed, involvement processes, levels of participation, and learning outcomes [3,4]. For co-design to work effectively, all stakeholders must be provided opportunities to equally engage and learn, and their values and culture must be incorporated into the development process. Failing to optimize co-design approaches across the development process inhibits the representation, learning and empowerment of stakeholders [2].

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Barki and Hartwick (1989) suggested that we need to rethink what it means to involve someone versus having someone participate. Involvement can be described as the extent to which a system reflects what is important and relevant to the stakeholder, versus participation can be described as the extent to which specific actions or behaviors are employed by the stakeholder during system development [5]. The term involvement has many connotations ranging from soliciting end-user insight to involving them as co-designers.

Determining the value of the stages and levels of participation is further complicated when we define this based on the target population for this systematic review of those aged 60 or older. The World Health Organization (WHO) estimates that by 2050, the world population of those aged 60 and older will increase to 2 billion, or nearly double what it was in 2015 [6]. As the number of people within this age group increases, aligning systems that meet their needs is imperative. While user needs and experience with technology may vary at any age, system designers have focused on proven areas that address the accessibility needs when designing for the aging population [7,8], but less has been done to evaluate the learning outcomes associated with involving older adults as well as the levels of participation during the development of electronic healthcare tools (EHTs). Involvement and participation occur at varying levels, and when users are more fully engaged, they feel a stronger sense of participation, which has the potential to impact their learning outcomes [2]. While several factors make up the stages of involvement and participation, further evaluation as to the level of involvement at each stage is necessary, as little is known about which approaches are the most effective in involving older adults [9]. The intent to involve stakeholders can be difficult as the framework to conduct these efforts is not prolific [3]. This is further exacerbated by the hierarchical challenges that may arise between designers, healthcare providers, and older adults, which can lead to selective inclusion of ideas [7,10–12]. However, by addressing this challenge, the opportunity for mutual learning and knowledge transfer among stakeholders can occur [2,13–15].

EHTs were selected for this review because the evidence supports the significant impact these tools have on improving health outcomes, including increasing knowledge, improvement in risk perception, and improvement in communication between patients and their clinicians [16]. We are using the term EHT to encapsulate all terms classified and defined by the WHO Digital Health Interventions [17]. EHTs are interactive web applications, mobile applications or wearables that track or provide guidance on relevant healthcare issues to improve wellbeing. For instance, monitoring health risks (i.e., risk of falling, risk of heart failure), or personal health information management (i.e., patient decision aids, access to health records and educational materials). While evidence supports the value in involving older adults in the design process, and positive impacts have been reported in the utilization of EHTs, we need to be cognizant that inconsistencies exist within these approaches, therefore evaluating the involvement process further is necessary to ensure that the expended efforts of older adults result in a mutual exchange of knowledge and improvements to the development process without detriment to their well-being.

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In a previous systematic review by Fischer et al, evidence supported involving older users in technology design, as indicated in the positive outcomes of involvement, specifically regarding learning, adjusted design, and an increased sense of participation [18]. Further supported in the review are the comparison examples between outcomes of designs that involved older adults during the development of the tool versus those that did not, and of the results of those who involved older adults the developed tool was determined to better fit the needs of the intended population [18]. Fischer et al., explored the purposes and consequences of involving older adults in technology design, such as involvement, sense of participation, and learning. While we review similar consequences, we take this a step further and expand on the involvement processes and levels of participation. We also analyze learning as an outcome, including who benefits from the learning as well as the user testing measures employed. A previous systematic review published by Lancaster et al. focused on electronic health tools but synthesized the usability and effects [19] of the tools, rather than co-design approaches. Additionally, their target population was not defined as older adults. Another systematic review was published with similar methods and target populations as our review [1], however, they focused on assistive technology which promotes independence of living rather than on the development of EHTs. While similarities exist between our review and these, we have enough distinction from these reviews that our search will draw from a different group of studies, therefore will provide for a robust review of the current state of co-design approaches used to involve older adults in the development of EHTs.

Aim

Primary Research Question:

Synthesizes the current state of utilizing co-design approaches with older adults in the development of electronic healthcare tools.

Secondary Research Questions:

- What approaches were used for involving older adults in developing EHTs and how were they defined?
- What theoretical frameworks and design principles were used to develop EHTs for older adults?
- Was iterative development utilized? If so, how many phases were conducted?
- What were the subjective and objective measures used for study endpoints, and were the measures validated?
- What involvement processes were employed?
- What were the level of participation for older adults?
- Did participants reflect on testing measures? (i.e., reflect on why an error was made)
- Was there a bi-directional flow of knowledge between participants and researchers?

Methods

Patient and Public Involvement

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No patient involved

Systematic Literature Search Strategy

We conducted a systematic review, following the Preferred Reporting Items for Systematic Reviews (PRISMA) 2020 checklist [20]. Consultations with an experienced research librarian (JB) were held to develop our search strategies. In March 2021, searches were developed for PubMed, Embase, and Scopus. A broad search strategy was utilized to ensure all co-design approaches that involve end users in the design of EHTs were captured in this review as depicted in (Supplementary Appendix A-Search Results).

The search results were screened according to the criteria defined in Table 1.

(Table 1 - Inclusion and Exclusion Criteria)

Data Extraction and Synthesis

Data extraction (selection and coding)

All records were uploaded to Covidence for screening purposes [21]. The authors (AC, KA) independently screened titles and abstracts that were identified in the search results from the database and excluded those which did not meet the selected inclusion criteria. A third reviewer (LM) resolved conflicts. After acquiring the full-text article, each author (AC, KA) independently reviewed and assessed each article to determine if it met the inclusion criteria. The final list was reviewed by the two authors to confirm which articles should remain in the review. A third reviewer (LM) resolved conflicts.

Study data were extracted for analysis and included study setting, population characteristics (age and health status), theoretical frameworks and principles used, terms used to define co-design approaches, types of electronic healthcare tools, user testing measures, types of co-design involvement processes (workshops, focus groups, interviews, prototyping, think-aloud, observation, contextual inquiry, usability testing, etc.), levels of participation, and learning outcomes.

Risk of bias (quality) assessment

While our systematic review was not focused on reported outcomes, and may not have required a quality assessment, as suggested in similar systematic reviews related to involvement processes, we wanted to provide qualifications as to how we assessed each study [18,22]. We evaluated the quality of studies included in this review, using the Agency for Healthcare Research and Quality (AHRQ) Evidence-based Practice Center approach [23]. Per this approach, two authors (AC and KA) rated studies on directness, consistency, limitations, and reporting bias, and a consensus was reached through a discussion of conflicts [23].

Strategy for data synthesis

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The data from each study were combined to provide a narrative or descriptive overview. We developed a data extraction form in Excel using a random sample of 2-3 including full texts of articles and revised it iteratively. The main characteristics extracted from the included studies are outlined in (Supplementary Appendix B-Study Classification Methods). A comprehensive assessment of all study results has been conducted, highlighting similarities and differences in each study regarding the types of co-design approaches utilized. This comparison was necessary, as there was anticipated heterogeneity in the terminology and approaches. We used widely reported classification methods to categorize the included studies on the types of healthcare tools [17], types of involvement processes [24], levels of participation [3], and levels of learning [25,26].

To determine whether the tool described in the study met the inclusion criteria, each tool was classified using the WHO Classification of Digital Health Interventions, which is a framework that focuses on objectives within health sectors and categorizes the digital technology that fits within each sector [17]. The classification scheme focuses on interventions for 4 primary users, including clients, healthcare providers, health system or resource managers, and data services. Our target population is older adults, therefore they fit within the client framework, but it should be noted that caregivers also fit within this grouping. The WHO Classification framework was developed to synthesize evidence for digital health interventions, with the terminologies and definitions refined through public feedback, therefore is an appropriate framework for categorizing the EHTs in this review [17].

The classification tool for clients consists of 7 categories, each further broken down into subcategories. The tools which were developed or redesigned within each included study have been classified within 5 of the 7 overarching WHO categories, including Targeted Client Communication, Client to Client Communication, Personal Health Tracking, Citizen Based Reporting, and On-Demand Information Services to Clients.

Involvement processes were extracted from the studies and classified by the co-design framework originally developed by Leinonen and since further refined, representing 4 distinct phases, including contextual inquiry, participatory design, product design, and software prototype as hypothesis (functional prototypes) [24,27]. Levels of participation were extracted and classified based on Vaughn's framework, with levels ranging from being informed, consulted, involved, in collaboration as a co-leader, to empowering oneself and others [3]. Table 2 provides Vaughn's definitions for levels of participation. Leinonen and Vaughn's framework were combined to evaluate how the core involvement processes that span across the various co-design approaches relate to the five levels of participation, as depicted in Figure 1.

(Table 2 - Levels of Participation)

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(Figure 1- Extracted from Vaughn's Involvement Framework and Leinonen's Process Framework and modified to reflect the bi-directional aspect of involvement as a process in collaboration with the stages of the co-design involvement processes.)

Within Leinonen's framework, participatory design is defined as the stage at which stakeholders have the opportunity to provide input [27]. For this literature review, this subcategory encompasses all involvement processes extracted from the studies in which stakeholder insight was represented. Other involvement processes were defined but did not fit within the cataloging of processes, which included the use of eye-tracking [8,28], teach-back methods [29], and spending time teaching older adults how to use technology in a classroom setting to better prepare them for engaging in the development process [14]. While the latter two methods do not fit within the cataloging of processes, the extent to which this involvement occurred provided a higher level of participation and associated learning outcome, suggesting that participants were empowered through these methods.

Learning outcomes were extracted from each study and classified using both Bateson's levels of learning, and Argyris' mutual learning model [25,26]. Bateson's learning framework consists of five learning levels, however only Learning Levels 0 –2 are applicable to this review [25,30,31]. Learning Level 0 refers to when a participant is provoked to respond but where no change to their action takes place. Learning Level 1 is an engaged response, in which a participant's errors are reflected on and corrected. Learning Level 2 utilizes context and tacit knowledge to draw insights for higher level learning to take place. Argyris' model consists of single and double loop learning. Single loop learning (synonymous with Learning Level 1-Reflective Learning) is the ability to detect and correct an error, which is the result of an ineffective action, and double loop (synonymous with Learning Level 2-Mutual Learning) occurs when significant actions are evaluated by the extent to which they help participants generate valid and useful information [26]. Learning Level 2-Mutual Learning is where participants, designers and researchers learn from each other, or when participants learn from reflecting on their own efforts and errors during user testing [2], including when participants are taught or exposed to technology prior to user testing [14,29]. As this review focuses on the development of EHTs, specifically the tools used to inform healthcare decisions, evaluating the role of learning through co-design approaches is necessary. If learning does not occur to the extent necessary to make qualified decisions, the information generated for consumption by the end user may not be validated.

Results

The initial database search returned 835 articles, including 495 from PubMed, 246 from Embase, and 94 from Scopus, of which 210 were duplicates, leaving 625 articles for the title and abstract review. Upon completion of the title and abstract review, 156 articles remained for a full-text review. The further assessment resulted in 25 studies that met the inclusion criteria. The review process is depicted in the PRISMA flow diagram [32] in Figure 2.

(Figure 2 – PRISMA Flow Diagram)

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Study Quality Assessment

All included studies were found to be of high quality, with no concerns regarding directness, consistency, precision, and reporting bias. We did find some limitations, which we rated as "medium" level, including two studies that did not indicate frameworks [8,33], and six of the twenty-five studies had limitations due to sample size [8,11,34–37]. Overall, we did not find any studies that would rate as a "high" level limitation.

Characteristics of Included Studies

Table 3 summarizes the characteristics of all included studies. Twenty-five eligible studies were identified from our initial database search. While all studies included older adults over the age of 60, we were unable to define the overall mean age, as 6 studies simply provided the age range [7,29,37–40], and 3 simply stated participants were 65 or older [11,12,14]. Overall, the studies included participants in the age range from 60 – 91, and of the studies which provided specific ages, we have a median age of 70.9. Of the 25 studies, 13 studies involved additional participants including caregivers, healthcare providers, and others such as researchers [41,42], department heads & policymakers [42], and control group participants [43]. A total of 9 studies targeted participants with specific diagnoses (Table 3).

Of the 25 studies, the number of stakeholders across all studies ranged from 7 – 135 participants, with a mean of 35.5 for participants classified as older adults, and a mean of 40.8 when including all stakeholders.

The settings and locations varied by study. Of the 25 studies, only 11 reported a specific setting in which the study was held, including laboratories, clinics, homes, community and senior centers, as well as remote sessions via Zoom. 50% of the studies took place in North America, 46% in Europe, and only one study (4%) was from Asia.

(Table 3- Summary Characteristics)

Co-Design Approaches

Table 3 includes co-design approaches, number of phases conducted in each study and whether they defined their approach. Overall, 9 different co-design approaches were represented among the 25 studies. Of the 25 studies, 7 studies indicated the use of more than one co-design approach [8,9,37,39,41,43,44], and 2 studies stated that stakeholders were involved in development but did not indicate a specific approach [28,45]. Overall, 19 of the 25 studies conducted their study in multiple phases, ranging between 2 to 6 phases, and an overall mean of 3.4 phases per study. Of the 6 studies which did not employ iteration in development, 2 labeled user involvement as either a human factors or usability evaluation approach and did not provide definitions to the approach utilized [33,43]. Reviewing the definitions of co-design approaches, 8 studies refer to involvement with end-user/stakeholder, 5 studies refer to system usability, 34 studies refer to user needs, 3 studies refer to using prototypes, and 3 studies refer to iterative development. Studies did not determine which involvement processes were used when defining the approach.

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Theoretical Framework/Co-Design Principles

Studies did indicate that specific frameworks or principles were utilized to guide them in selecting which involvement processes to follow. Supplementary Appendix C summarizes the key theoretical frameworks or co-design principles employed for each study. Studies that reference the same terminology for both the approach and framework such as Human-Centered Design (HCD), did not use iteration in their definition, but referenced standards such as HCD Methodology – ISO 9241-210 which specifically states this is an iterative process [46].

Of the 25 studies, 2 did not indicate that their study was guided by a particular framework or principle [8,33], although one study was defined as using co-design with a participatory experience, and the other as using human factors, neither provided clarification on their approaches. As 18 different frameworks or design principles were represented across the 25 studies (Supplementary Appendix C), it is difficult to conclude which involvement processes should be included within each framework. The most commonly reported frameworks were Human Centered Design Methodology (ISO Standards) (n=5), International Patient Decision Aid Standards (IPDAS) (n=4), and User Center Design (n=3). There are inconsistencies among the studies which indicated utilizing similar frameworks or principles, such as HCD, UCD, and IPDAS, which warrants further exploration into what it means to state a study is employing a particular framework or principle. Interestingly, 6 of the 25 studies developed recommendations for frameworks that could be employed when collaboratively designing EHTs with older adults.

Electronic Healthcare Tools (EHTs)

Types of Electronic Healthcare Tools (EHTs)

Table 4 summarizes the EHTs based on the WHO Classification of Digital Health Interventions. In 12 of the 25 studies, the EHT that was developed or redesigned fits within multiple categories, therefore, was cataloged in multiple categories. The overall objective was to ensure the tools included in the selected studies fit within a categorization that could be defined as those which support healthcare decisions, which was achieved through this classification process.

(Table 4 –WHO Classification of Digital Health Interventions)

Subjective and Objective Measures

We extracted 18 different measures, including 13 subjective, 3 objective, and 2 physiological measures, accounting for 19 of the 25 included studies, as indicated in Table 5. These studies utilized between 1 to 6 measures, with a mean of 3.2 measures per study, with usability as the most common endpoint, being measured in 13 of the included studies. Of the 19 studies that utilized these measures, only 9 studies used objective measures but were focused on three specific endpoints (errors, performance, and need for assistance). Within the 18 studies that used subjective measures, we extracted 23 different endpoints.

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We further categorized these measures to evaluate whether studies were using validated measures and found that only 5 studies (20% of all included studies) used validated subjective measures to evaluate their EHT. Other than the SUS scale, which was utilized in 20% of the studies to measure usability, the other validated subjective measures identified were only used in 4% - 8% of studies, while 12 distinct non-validated Likert-scale assessments spanned across 52% of all studies. Usability was also the most specified measure in non-validated Likert-scale assessments, taking place in 8 studies. Overall, 13 of the 25 studies reported measuring usability, but less than half used a validated measure.

(Table 5 – User Testing – Subjective and Objective Measures)

Stakeholder Involvement

Types of Involvement Processes

A variety of involvement processes were used in the 25 studies, as indicated in Table 6, which reports 152 processes across all the included studies. Supplementary Appendix C includes the full list of involvement processes used within each study. The involvement processes that were most widely reported were interviews and prototypes (medium fidelity) (n=18), followed by prototype (functional) (n=15) and the least reported was remote evaluation (n=1). Each study utilized between 2 to 11 involvement processes, with a mean of 6 processes per study.

(Table 6 – Co-Design Involvement Processes by Learning Levels)

Levels of Participation

Defining each study based on the level of participation presented challenges. Each study was assessed to have reached a participation level based on Vaughn's definitions [3] and cross-classified by co-design approach (Supplementary Appendix D). While Vaughn's definitions are related to research processes, they can be translated to the levels of participation in developing EHTs. The challenge in mapping each level against the co-design approach is that the studies did not define distinct points at which participants were involved. However, we can see that most studies employed the first three levels of participation, that of inform, consult, and involve, as evidence suggests the information provided to the research team was valued and assessed for inclusion into various design elements of the tool. However, fewer studies indicated that participants had an equal level of decision-making when making iterative changes to the design, and even fewer empowered the participants to take a lead in engaging others on the usefulness of the tool.

Figure 3 depicts each level of participation and the percent breakdown of associated involvement processes. The most distinct comparison is between the studies classified as Level 5-Empower (No) versus Empower (Yes). This comparison reflects a decrease in participatory design (from 69% to 55%) and an increase in functional prototypes (from 2% to 14%), respectively.

(Figure 3 - Involvement Processes (Percent of Total))

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Learning Outcomes

Table 3 summarizes the learning levels of each study. All 25 studies were categorized as having met learning Level 0, in which researchers benefitted from participants' responses to specific user testing measures. While 8 studies met Learning Level 1, in one of the 8 studies, only a subset of the participants were involved in reflective learning [36] as this occurred in later phases of the study, in which the number of participants had decreased by more than half. In addition to these 8 studies, one study acknowledged to the participant they had committed an error but did not provide feedback [43] and another study acknowledged it as a study limitation that they did not review results with participants [40]. 7 studies reached Learning Level 2-Mutual Learning, utilizing transference of knowledge that focused on prior experience with the technology. This benefitted both researchers and participants, indicating a bi-directional flow of knowledge occurred. It should be noted that these 7 studies were also classified as having reached the empower participation level.

Table 6 summarizes the difference in the types of involvement processes used when studies engaged researchers and participants in mutual learning. While additional studies engaged in observational activities that provided opportunities for additional context, these studies did not engage in mutual learning. For example, an additional 2 studies utilized contextual inquiry, however, in these studies the learning was unidirectional and only benefitted the researchers.

Discussion

Key Findings

This review focused on synthesizing the current state of involving older adults in co-design approaches to develop EHTs and aimed to identify how the co-design approaches utilized mutual learning techniques to benefit older adult participants. We extracted evidence from each study to categorize the terms used to describe co-design approaches, as well as to draw themes around how these terms were defined. We identified the EHTs and categorized them by health sector classifications as well as the subjective and objective measures employed. This review also explored the frameworks and principles that determined the stakeholder involvement, including types of involvement processes, levels of participation and learning outcomes.

Co-Design Approaches

As anticipated, heterogeneity exists within the definitions used to describe co-design approaches. While 13 definitions were represented across the studies, general themes presented suggest that the approaches could be used interchangeably. While this may further exacerbate a problem, and lead to inconsistent use of terminology, what is of notable importance within this review, is not so much on the name of the approaches used, but how they are interpreted and then employed to determine the methods for involving and co-designing with older adults. Not surprisingly, each study specified either definitively within their

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aim or subsequently, that end-users should have some level of involvement or participation during the development life cycle process. However, studies did not specify the extent to which older adults should be involved, nor did they specify whether a learning model should be employed. Of interest, is that only 36% (n=9) of included studies had a mutual exchange of knowledge between the research team and study participants, even though all studies were classified as having utilized a co-design approach. For the remaining studies (n=16; 64%), participants engaged in processes in which they simply provided feedback on the EHT and did not engage in processes that facilitated a mutual exchange of knowledge. Future co-design studies involving older adults should utilize improved co-design frameworks and optimize mutual learning.

Frameworks and Principles

We chose not to cross-classify this variable as to not misrepresent relationships; however, the diversity of frameworks extracted exemplifies a need for a framework that is specific to engaging older adults in the development of EHTs, which can be consistently utilized, and which also provides validation to the involvement processes that should be employed. This need is further exemplified as 25% (n=6) of the included studies suggested recommendations for a framework that could be used when engaging older adults in the development of EHTs. Further research is needed to determine whether the proposed, or a combination of existing and proposed frameworks can be validated to optimize older adults' involvement, one that provides an environment for mutual learning and empowerment.

Electronic Healthcare Tools (EHTs)

While the EHTs that were applied in the studies spanned multiple categories under the WHO classification, the majority focused on tools that provided the ability to search for information on a particular health topic or that allowed participants to track their health information through self-monitoring, including accessing personal health records, use of wearables, or documenting health information. Each study involved participants in developing an EHT that would support older adults' healthcare decisions. Previous studies have reported positive impacts to both older adults and improvements in technology when engaging them in the development lifecycle [1,2].

Furthering the notion that a framework, specific to older adults in developing EHTs, needs to be evaluated, is concerning the inconsistency of user testing assessments. We extracted evidence that 19 different measures were utilized in 76% (n=19) of the included studies and found that only five studies (20% of all included studies) used a validated subjective measure to evaluate their EHT. Usability was the most specified measure, and while the SUS, a validated usability assessment was utilized, more than half of the studies utilized a non-validated Likert-scale assessment. This discrepancy in assessments for user testing makes it difficult to draw definitive conclusions as to whether studies are consistently measuring what they intended to measure. This is consistent with a recent review that found studies often use non-validated

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questionnaires rather than readily available validated assessments and that despite recommendations to measure learning, this is rarely employed [47].

Stakeholder Involvement, Participation and Mutual Learning

While we found inconsistency in the involvement processes among the studies, over 70% included interviews and prototyping, others included a range of processes such as cognitive walkthroughs, think-aloud sessions, surveys, scenarios, and focus groups, and only 20% of the studies utilized contextual inquiries and use cases. While fewer studies utilized contextual inquiries, the studies that employed this process were more likely to be categorized as having both empowered users and involved them in mutual learning. For studies in which participants and researchers benefitted from mutual learning, we observed a higher utilization of specific involvement processes including contextual inquiry, cognitive walkthroughs, focus groups, working group workshops and functional prototyping, and a decrease in think aloud sessions. This is consistent with previous research by both Bateson and Argyris as well as recent studies that evaluated learning outcomes [30,48] and suggests the importance of understanding user context and involvement in group activities that facilitate reflective and mutual learning. While Argyris also suggests that when groups work on important topics, they are less receptive to corrective feedback, inhibiting learning and potentially producing information of lower value; we observed that these processes were effectively employed and solicited valid information on important topics, or that of improving the development of EHTs that are meant for healthcare decisions.

To the best of our knowledge, current frameworks, or principles have not provided guidance on distinguishing the levels of participation, nor the validated involvement processes that should be utilized when engaging older adults in developing EHTs. To provide structure to this notion, this review drew upon Leinonen's framework for processes and Vaughn's framework for levels of participation. After extracting evidence of both variables from the 25 studies, and overlaying these on top of the two frameworks, we began to recognize the areas in which we need to further explore. We need to explore what it means to involve older adults, specifically which involvement processes are more likely to lead to empowerment and mutual learning. This is important because the evidence extracted from the studies suggests that bi-directional knowledge, which benefits both older adults and researchers, may increase as participation levels increase.

It has also been suggested that exposure to the technology before usability testing has shown to be an effective means of eliciting users' needs, such that the increased knowledge and awareness of the tool inspires ideation and innovation [9,37], and that tacit knowledge or real-life context increases the ability to portray specific user needs [42]. However, there are concerns about bias if an individual is exposed to technology before soliciting their design ideas [49]. Not to dismiss the concern for bias, but further evidence supports the notion that experienced users of technology can provide more meaningful insight during design sessions and could provide feedback more representative of the target population [50]. We also see that

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allowing users to engage with the technology for an extended time or ensuring they understand how to use the technology, reduces frustration when contributing to design ideas [14,29,49]. For example, when older adults were asked to participate in two similar design activities, they were more involved when they had familiarity with the topic, compared to being asked to innovate on a novel topic [34].

Suggestions have been made to separate co-design processes from outcomes and focus on the role that participants have to be co-designers and co-initiators [51], which supports the values defined by the fourth and fifth levels of involvement, that of collaborate and empower, respectively. We should be looking at involvement as a process, one in which through various stages of the development life cycle, users have the potential to cross through the spectrum of involvement highlighted in Vaughn's framework. By combining Leinonen and Vaughn's framework, we can see how the involvement processes that span across the various co-design approaches, can be used in iterative development, but through these various stages, users can participate in the processes at different levels, from being informed, consulted, involved, in collaboration or as a co-leader, to empowering oneself and others. This is consistent with a review by Fischer et al. that also suggest that involvement as a process should be a consideration for future research and policy, specifically the extent to which users are involved and receive benefit [18]. Further, we suggest that additional exploration into the extent to which a greater understanding of the tool can contribute to richer feedback is necessary [9]. We also suggest the levels of participation must be bi-directional for mutual learning to occur, as depicted in Figure 1, and when realized within the development lifecycle will result in greater contributions to the development of the tool.

Limitations

As our inclusion criteria spanned an array of terms that we categorized as co-design approaches, we had several approaches that were only represented by one study, which may not fully represent all studies that incorporate such approaches. One of the biggest limitations of this study was the difficulty in extracting and classifying the information, however we used mitigation strategies (e.g., established frameworks and classification methods) to support the review of variables including types of EHTs, stakeholder involvement and participation and levels of learning. Studies that employed co-design approaches were included in our review, however, we limited our review to the research groups' initial study and did not evaluate whether subsequent papers by these groups provided a broader analysis of user experience.

Conclusions

This review depicted not only the heterogeneity in the terminology used to define co-design approaches, but also the gaps in frameworks and design principles for developing EHTs for older adults. Most studies in this review developed EHTs specific to patient engagement and designed their study to include older adults' feedback in the design, indicating a collaborative focus centered on meeting the needs of this population. However, we found that studies did

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not consistently utilize validated assessments for conducting usability testing. We also found gaps in the evidence to support which involvement processes should be used when co-designing with older adults. Additionally, gaps exist around which learning models should be employed when conducting user testing with older adults. Future research needs to evaluate how empowerment is associated with mutual learning, whether this bi-directional knowledge can occur at all participation levels, and whether the benefit to both older adults and researchers increases as the levels of participation increases. To ensure studies are optimizing the involvement processes, participation and learning of older adults when developing EHTs, there is a need to place more emphasis on the approaches promoting mutual learning.

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Contributors

AC and KA conceived of the broad concept for the literature review. AC developed the protocol, then worked with KA and JB to develop a search strategy. AC and KA independently screened titles and abstracts that were identified in the search and excluded those which did not meet the selected inclusion criteria. LM resolved conflicts. After acquiring the full-text articles, AC and KA independently reviewed and assessed each article to determine if it met the inclusion criteria. The final list was reviewed by the AC and KA to confirm which articles should remain in the review. A third reviewer LM resolved conflicts. AC extracted and synthesized the data. AC wrote the review, with meaningful insight in the structure of the analysis and conclusion with KA and LM. Final review of the manuscript by KA, LM, AK, and DR.

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Competing Interests

None declared

Patient consent for publication

Not required

Ethics Statement

This is a systematic review, without human involvement, as such an ethical assessment was not required.

Data availability statement

All data relevant to the study are included in the article or uploaded as supplemental information.

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Figures

Figure 1- Extracted from Vaughn's Involvement Framework and Leinonen's Process Framework and modified to reflect the bi-directional aspect of involvement as a process in collaboration with the stages of the co-design involvement processes.

Figure 2 – PRISMA Flow Diagram

Figure 3 - Involvement Processes (Percent of Total)

Tables

Table 1 - Inclusion and Exclusion Criteria

Selection Criteria	***Inclusion Criteria***	***Exclusion Criteria***
Participants	Older adults aged 60 or older, without regard to gender, race, or ethnicity	If the study population included adults of all ages, including greater than 60 years of age, but evidence could not be extracted from those greater than 60 years of age, the study was excluded
Study Setting	All study settings were included	
Study Design	All study designs were included	
Methods and Tools	Studies published that employed co-design	Studies that used non-electronic healthcare tools

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	approaches in the development of electronic healthcare tools	
Language		Studies in which publications are not in the English language were excluded

Table 2 - Levels of Participation

Levels of Participation	***Definitions-Vaughn et al. [3]***
Inform	Information is shared with community members and could become more participatory if members ask for information relative to their interests.
Consult	Participants provide feedback to the researchers, as related to a specific decision point.
Involve	Participants provide feedback to researchers during the entire process.
Collaborate	Participant input is valued equally, as a co-leader in the development process.
Empower	Participants take a key role in leading the initiative to get others engaged.

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Table 3- Summary Characteristics

First Author's last name	Year of Pub	First author's location	*Digital Health Intervention	Co-Design Approach	Co-Design Approach Defined	No. of Phases	Learning Level	Learning Beneficiary	No. of Stakeholders	No. of OA	OA/P	Ca	HP	Ot	Age of older adults (mean)	Diagnosis
Abujarad [52]	2021	USA	PHT CBR ODISC	User Centered Design	Yes	-	Level 0	Research Team	38	24	X	X	X		69	
Ahmed [34]	2019	USA	TCC PHT	Participatory Design	Yes	-	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	9	7	X	X			67	Heart Failure
Bogza [29]	2020	Canada	PHT ODISC	User Centered Design	No	6	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	19	12	X	X	X	X	(2) 60 – 64 years (6) 65 – 74 years, (3) 75 – 84 years, (1) >85 years	Mild Cognitive Impairment (MCI)
Cornet [12]	2020	USA	PHT ODISC	User Centered Design	Yes	3	Level 0	Research Team	63	50	X	X	X	X	>65	Heart Failure
Gonzalez [33]	2014	USA	PHT	Human Factors	No	-	Level 0	Research Team	12	12	X				77.7	
Grimaldi [11]	2020	Italy	ODISC	Human Centered Design	Yes	4	Level 0	Research Team	10	10	X				> 60	
Gustafson [14]	2016	USA	CCC	User Centered Design	Yes	5	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	135	135	X	X	X	X	> 65	
Harrington [9]	2018	USA	PHT	Co-Design Human Centered Computing Interaction Design Process Participatory design	No	2	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	39	39	X				72.1	
Harte [7]	2017	Ireland	PHT	Human Centered Design	Yes	3	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	12	12	X				61-85	

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First Author's last name	Year of Pub	First author's location	*Digital Health Intervention	Co-Design Approach	Co-Design Approach Defined	No. of Phases	Learning Level	Learning Beneficiary	No. of Stakeholders	No. of OA	OA/P	Ca	HP	Ot	Age of older adults (mean)	Diagnosis
Hoffman [39]	2020	USA		Co-Design User Centered Design	No	3	Level 0	Research Team	16	8	X	X			Phase 2: 65-84 Phase 3: 61-89	
Holden [53]	2020	USA	ODISC	User Centered Design	Yes	3	Level 0	Research Team	23	23	X				67.6	
Kim [38]	2018	USA	PHT ODISC	Participatory Design	No	2	Level 0	Research Team	17	17	X				65-80	
Mansson [35]	2020	Sweden	PHT	Co-creation	Yes	2	Level 1 Sgl Loop	Research Team Participant	10		X				76	
Martin-Hammond [36]	2015	USA	ODISC	User Centered Design	Yes	3	Level 0	Research Team	7	7	X				70	
Nguyen [42]	2019	Netherlands		Co-Design	No	3	Level 0	Research Team	56	56	X			X	68.9	
Nielsen [44]	2018	Denmark	TCC CCC PHT ODISC	Participatory Design Co-Design	Yes	3	Level 0	Research Team	54	36	X	X	X		70.3	Hearing impairment
Or [43]	2012	Hong Kong	TCC PHT	Usability Evaluation Approach Usability Test Design	No	-	Level 0	Research Team	57	50	X			X	71.6	One of the following: hypertension, diabetes, heart disease, asthma, prostatitis, hypotension
Petersen [45]	2019	Netherlands	ODISC	Does Not State	No	3	Level 0	Research Team	71	32	X		X		70	Larynx cancer
Poirier [28]	2019	USA	ODISC	Does Not State	No	-	Level 0	Research Team	45	45	X				72.2	
Portz [40]	2020	USA	PHT ODISC	Human Centered Design	Yes	3	Level 0	Research Team	81	31	X	X	X		66-91	Heart Failure (palliative care patients)
Scandurra [37]	2013	Sweden	TCC CCC ODISC	Participatory Design User Centered Design	No	5	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	8	8	X				65-80	

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First Author's last name	Year of Pub	First author's location	*Digital Health Intervention	Co-Design Approach	Co-Design Approach Defined	No. of Phases	Learning Level	Learning Beneficiary	No. of Stakeholders	No. of OA	OA/P	Ca	HP	Ot	Age of older adults (mean)	Diagnosis
Ummels [41]	2020	Netherlands	PHT	Co-creation User Centered design	Yes	4	Level 1 Sgl Loop Level 2 Dbl Loop	Research Team Participant	63	63	X			X	70.3	
Van De Dijk [54]	2013	Netherlands	PHT	User Centered Design	Yes	-	Level 0	Research Team	86	86	X				72.5	COPD
Van Velsen [55]	2019	Netherlands	CBR ODISC	Participatory Design	No	2	Level 0	Research Team	82	82	X	X			71.5	
Wikberg-Nilsson [8]	2018	Sweden	ODISC	Co-Design Participatory User Experience Design	No	5	Level 1 Sgl Loop	Research Team Participant	7	7	X				75	Reduced hearing, eyesight, mobility, sensibility, and loss of memory function

*TCC = Targeted Client Communication, CCC = Client to Client Communication, PHT = Personal Health Tracking, CBR = Citizen Based Reporting, ODIFC = On-Demand Information Services to Clients

**OA/P = Older Adults/Patients, Ca = Caregivers, HP = Healthcare Providers, Ot = Others

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Table 4 –WHO Classification of Digital Health Interventions

WHO Classification of Digital Health Interventions	***Total studies (n=25), n(%)***	***References***
1.1 Targeted Client Communication		
1.1.2 Transmit targeted health information to client(s)	1 (4%)	[44]
1.1.3 Transmit targeted alerts and reminders	2 (8%)	[37,43]
1.1.4 Transmit diagnostics result, or availability	1 (4%)	[34]
1.3 Client to Client Communication		
1.3.1 Peer Group for Clients	3 (12%)	[14,37,44]
1.4 Personal Health Tracking		
1.4.1 Access by client to own medical records	3 (12%)	[34,35,38]
1.4.2 Self-monitoring of health or diagnostic data by client	7 (28%)	[7,33,34,41,44,52,54]
1.4.3 Active data capture/documentation by client	8 (32%)	[9,12,29,34,35,39,40,43]
1.5 Citizen Based Reporting		
1.5.1 Reporting of health system feedback by clients	2 (8%)	[52,55]
1.6 On-Demand Information Services to Clients		
1.6.1 Client Look-up of health information	16 (64%)	[8,11,12,28,29,36–40,42,44,45,52,53,55]

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Table 5 – User Testing – Subjective and Objective Measures

Measures	Endpoint	Total studies (n=25) n (%)	References
Validated			
Subjective Measures			
ACE (Altarum Consumer Engagement)	Patient engagement	1 (4%)	[34]
ASQ	Efficiency	1 (4%)	[7]
CES	Confidence & Computer Efficacy	1 (4%)	[52]
Decisional Conflict Scale (values clarity)	Values Clarity	2 (8%)	[28,29]
I-PANAS-SF	Emotional Reactions	1 (4%)	[52]
Likert scale (specified validity)	Comprehension/Recall, Graphicacy, Numeracy	1 (4%)	[28]
NASA-TLX	Cognitive Workload	2 (8%)	[7,12]
NVS	Health Literacy	2 (8%)	[12,34]
Ottawa Acceptability Scale	Quality	2 (8%)	[29,39]
SURE scale	Values Clarity	1 (4%)	[39]
SUS	Usability	5 (20%)	[7,12,29,52,53]
Us.E 2.0	Quality	1 (4%)	[11]
Non-Validated			
Subjective Measures			
Likert scale (non-specified validity)[a]		13 (52%)	[7,9,11,12,14,29,33,37,41–45]
Objective Measures			
Mistakes Counted	Errors	7 (28%)	[7,11,12,39,42,43,53]
Time to Complete	Performance	4 (12%)	[7,28,43,53]
Times Request Assistance	Need for Assistance	2 (8%)	[43,53]
Physiological Measures (eye-tracking)			
Cognitive Processes	Usability	1 (4%)	[8,28]
Gaze Fixation	Usability	1 (4%)	[8,28]

Satisfaction, User Acceptance, Appearance, Comfort, Functionality, Reliability, Usability, Usefulness, Ease of Use, Comprehensibility, Effectiveness, Value of Information [a]

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Table 6 – Co-Design Involvement Processes by Learning Levels

■ Contextual Inquiry ■ Participatory Design ■ Product Design ■ Prototype ■ Other

(color-coded to reflect association in Figure 3)

	Involvement Processes	*Level 0 (n=25), n(%)*	*Single Loop Level 1 (n=8), n(%)*	*Double Loop Level 2 (n=7), n(%)*
***Contextual Inquiry** (Rapid Ethnography)	Shadowing in context including semi/unstructured interviews, observation, etc.	6 (24%)	4 (50%)	4 (57%)
Participatory Design	Survey	10 (40%)	3 (38%)	3 (43%)
	Interview	18 (72%)	7 (88%)	5 (71%)
	Cognitive walkthrough	8 (32%)	4 (50%)	3 (43%)
	Think aloud	11 (44%)	2 (25%)	1 (14%)
	Remote evaluation	1 (4%)	1 (13%)	1 (14%)
	Focus group	8 (32%)	4 (50%)	4 (57%)
	Task analysis	10 (40%)	3 (38%)	2 (29%)
	Scenarios	7 (28%)	2 (25%)	1 (14%)
	Prototype (low fidelity)	8 (32%)	4 (50%)	3 (43%)
	Working Group Workshop	9 (36%)	5 (63%)	3 (43%)
Product Design	Use Cases	4 (16%)	2 (25%)	1 (14%)
	User Stories	9 (36%)	2 (25%)	2 (29%)
	Prototype (medium fidelity/throw-away)	18 (72%)	8 (100%)	5 (71%)
	Heuristic evaluation	5 (20%)	2 (25%)	1 (14%)
Software Prototype as Hypothesis	Prototype (functional/Beta version)	15 (60%)	8 (100%)	7 (100%)
Other	Other	5 (16%)	3 (38%)	2 (29%)
Total		**152**	**64**	**48**

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*The goal of contextual inquiry is to understand the context and the preliminary design challenges. It involves understanding the user, what the design challenges are and why they are being solved. Leinonen et al. [24] proposed ethnographic methods and benchmarking the environment to have a complete understanding of the major design challenges. Contextual inquiry is used as a category to remain consistent with Leinonen et al.'s terminology, but we recognize that this category can be broadened into "rapid ethnography" which Leinonen recommended in his follow-up research contributions [56].

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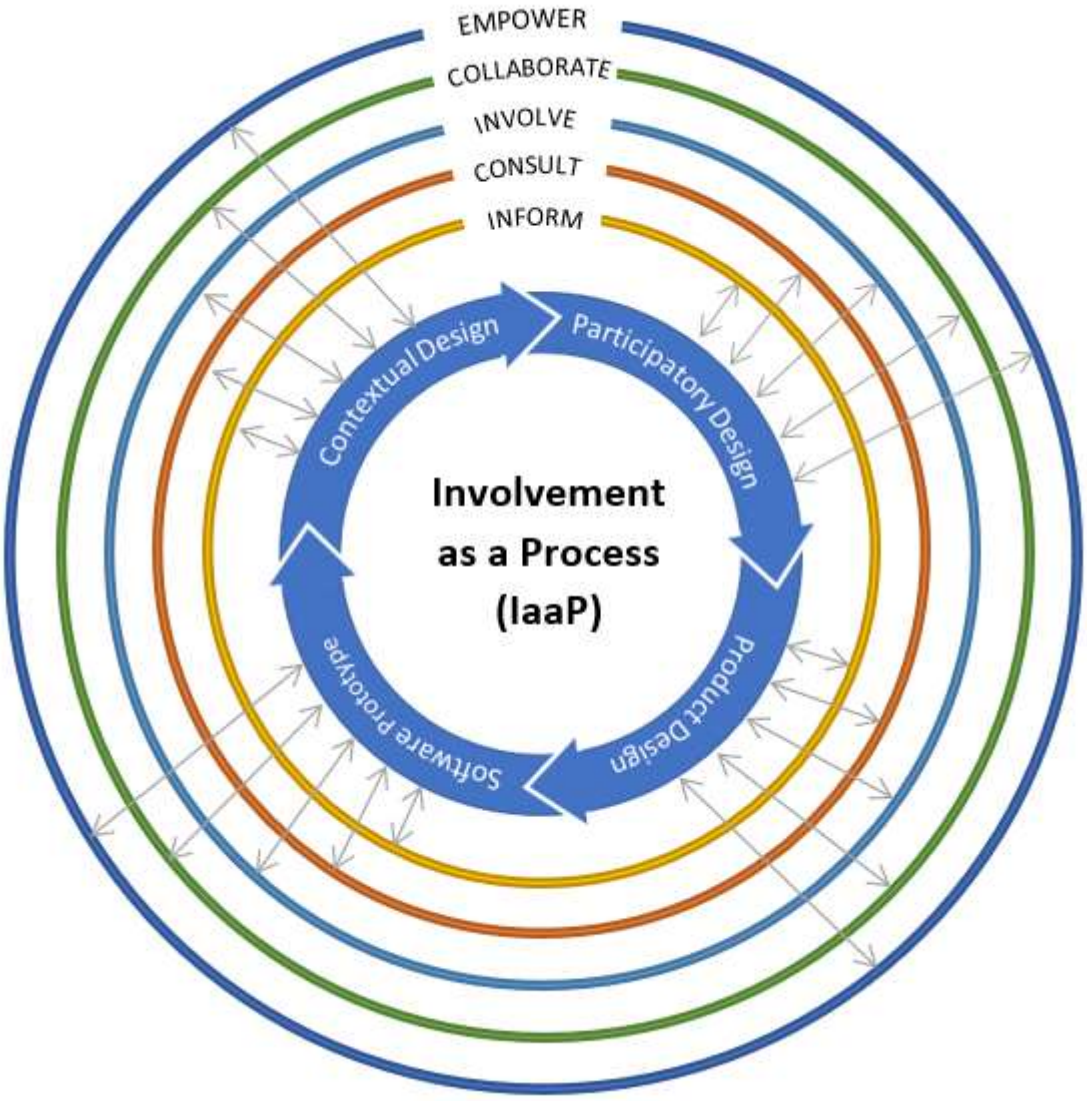


Figure 1- Extracted from Vaughn's Involvement Framework and Leinonen's Process Framework and modified to reflect the bi-directional aspect of involvement as a process in collaboration with the stages of the co-design involvement processes.

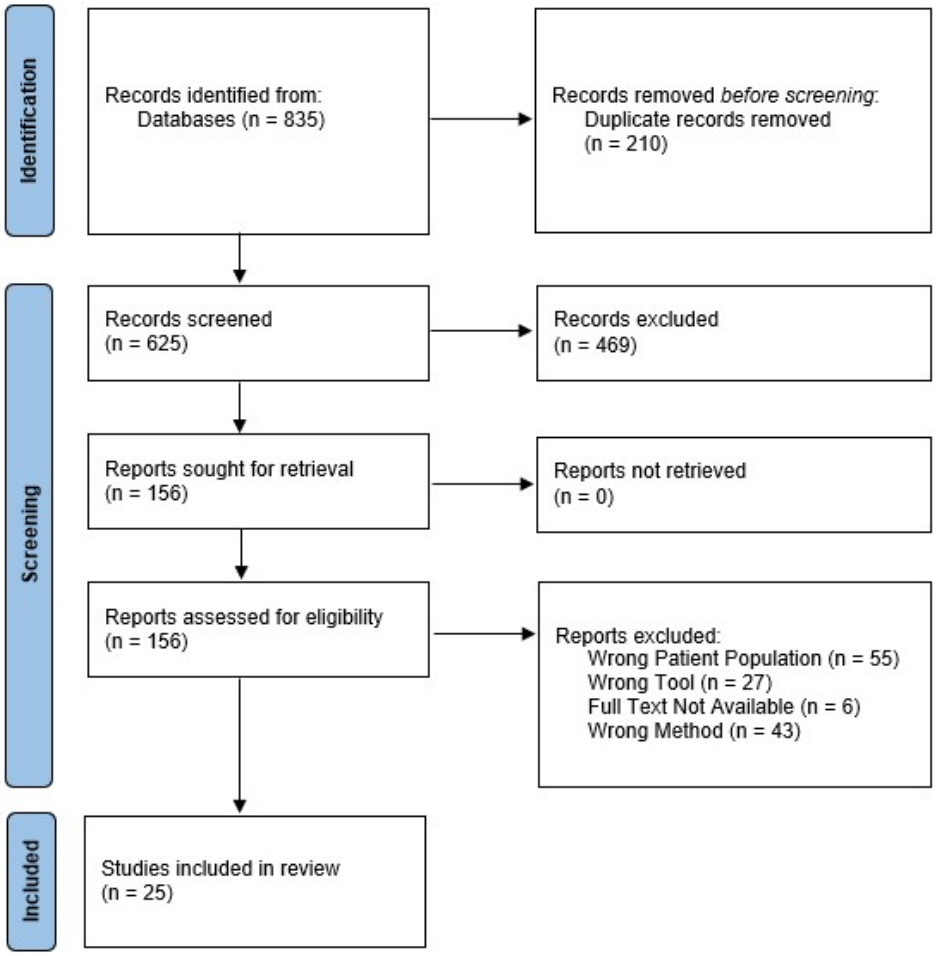


Figure 2 – PRISMA Flow Diagram

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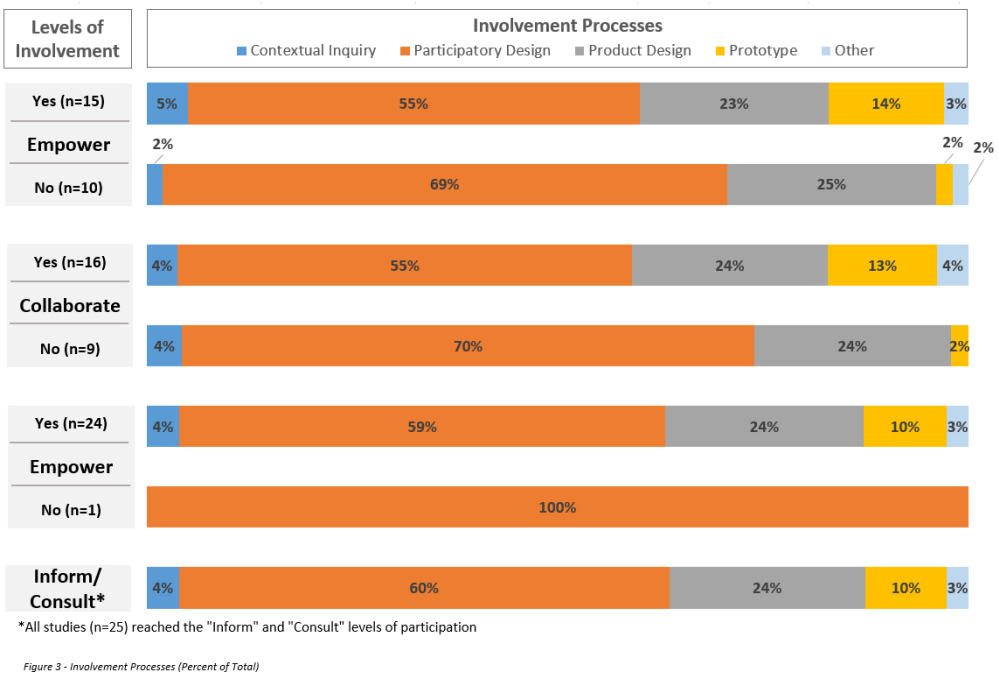


Figure 3 - Involvement Processes (Percent of Total)

Figure 3 - Involvement Processes (Percent of Total)

764x516mm (38 x 38 DPI)

Supplementary Appendix A - Search Results

PubMed – 3.17.2021		
Search #	Query	Results
#1	(Co-Design[tiab] OR Co-Creation[tiab] OR Co-Production[tiab] OR Participatory Design[tiab] OR Human Factors[tiab] OR User-Centered Design[mesh] OR User-Centered Design[tiab] OR Participatory Approach[tiab] OR Human-Centered Design[tiab] OR Universal design[mesh] OR Patient Participation[mesh] OR patient participation[tiab])	38,538
#2	(Aged[mesh] OR Older Adults[tiab] OR Older People[tiab] OR Elderly[tiab] OR Geriatric[tiab] OR Aged over 60[tiab] OR aged 60[tiab] OR Seniors[tiab] OR Over 65[tiab])	3,333,143
#3	(Electronic Health Records[mesh] OR Electronic Health Records[tiab] OR Electronic Health Record[tiab] OR Electronic Healthcare[tiab] OR eHealth[tiab] OR "mHealth"[tiab] OR Decision Aid[tiab] OR decision aids[tiab] OR Electronic Healthcare Aid[tiab] OR Healthcare Application[tiab] OR digital health[tiab] OR health information technology[tiab] OR health information technologies[tiab] OR health technology[tiab] OR health technologies[tiab])	59,271
#4	#1 AND #2 AND #3	495

Embase – 3.17.2021		
Search #	Query	Results
#1	('human factors research'/exp OR 'user-centered design'/exp OR 'Universal design'/exp OR 'Patient Participation'/exp OR ('co-design' OR 'Co-Creation' OR 'Co-Production' OR 'Participatory Design' OR 'Human Factors' OR 'user-centered design' OR 'Participatory Approach' OR 'Human-Centered Design' OR 'patient participation'):ab,ti)	42,464
#2	('aged'/exp OR ('older adult' OR 'older adults' OR 'older people' OR elderly OR geriatric OR geriatrics OR 'aged over 60' OR 'aged 60' OR 'over 60' OR 'aged over 65' OR 'aged 65' OR 'over 65' OR senior OR seniors):ab,ti)	3,460,578
#3	('electronic health record'/exp OR ('electronic health record' OR 'electronic health records' OR 'electronic healthcare' OR 'ehealth' OR 'mhealth' OR 'electronic decision aid' OR 'electronic decision aids' OR 'electronic healthcare aid' OR 'electronic healthcare aids' OR 'healthcare applications' OR 'Decision Aid' OR 'decision aids' OR 'digital health' OR 'health information technology' OR 'health information technologies' OR 'health technology' OR 'health technologies'):ab,ti)	59,924
#4	#1 AND #2 AND #3	246

Scopus – 3.17.2021		
Search #	Query	Results
#1	TITLE-ABS("co-design" OR "Co-Creation" OR "Co-Production" OR "Participatory Design" OR "Human Factors" OR "user-centered design" OR "Participatory Approach" OR "Human-Centered Design" OR "patient participation")	78,052
#2	TITLE-ABS("older adult" OR "older adults" OR "older people" OR elderly OR geriatric OR geriatrics OR "aged over 60" OR "aged 60" OR "over 60" OR "aged over 65" OR "aged 65" OR "over 65" OR senior OR seniors)	657,056

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#3	TITLE-ABS("electronic health record" OR "electronic health records" OR "electronic healthcare" OR "ehealth" OR "mhealth" OR "electronic decision aid" OR "electronic decision aids" OR "electronic healthcare aid" OR "electronic healthcare aids" OR "healthcare applications" OR "Decision Aid" OR "decision aids" OR "digital health" OR "health information technology" OR "health information technologies" OR "health technology" OR "health technologies")	54,356
#4	#1 AND #2 AND #3	94

- records before duplicates removed: 835
- records after duplicates removed: 625

Supplementary Appendix B - Study Classification Methods

Category	*Variable*	*Classification method*
Summary characteristics	Year of publication	Metadata
	First author's geolocation	Metadata
	Study setting	Metadata
	Number of stakeholders	Metadata, mean
	Number of older adults	Metadata, mean
	Study population	Metadata
Co-design approaches	Terminology or definitions	Manual, binary
	Theoretical framework/design principles	Manual, binary
	Iterative development	Metadata, mean
Stakeholder involvement	Participant types	Metadata
	Types of Involvement Processes	Leinonen et.al [24]
	Learning Levels	Bateson [25], Argyris [26]
	Levels of participation	Vaughn et.al [3]
Electronic healthcare tools	Types of healthcare tools	WHO classification of digital health interventions [17]
	User testing	Metadata

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Supplementary Appendix C

Author	Theoretical Framework/Co-Design Principle	Level of Involvement	Involvement Processes																User Testing Measures and Validation					
			A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	Endpoint	Other	Objective Measure	Subjective Measure	Physiological	Validated
Abujarad [52]	User Centered Design Approach	Empower	X	X	X	X	X			X					X	X		X		confidence and computer efficacy			CES		X
																				emotional reactions			I-PANAS-SF		X
																				usability			SUS		X
Ahmed [34]	Participatory Design Approach Developed recommendations for framework	Empower	X	X					X			X	X			X		X		health literacy			NVS		X
																				patient engagement - self report on health behaviors			ACE (Altarum Consumer Engagement)		X
Bogza [29]	Center for eHealth and Wellbeing Research guidelines Inductive thematic qualitative analysis approach IPDAS	Empower			X	X	X								X	X		X	X	quality of decision aid			Ottawa Acceptability Scale		X
																				satisfaction			Likart - unspecified		
																				usability			SUS		X
																							Decisional Conflict Scale		X
																				Comprehension	teach back method				
Cornet [12]	User Centered Design Developed recommendations for framework	Involve		X	X	X	X			X	X			X		X	X			cognitive workload			NASA-TLX		X
																				health literacy			NVS		X
																				usability			SUS		X
																				user acceptance			Likart - unspecified		
																				errors		Not stated			
Gonzalez [33]	none	Involve			X											X				appearance			Likart - unspecified		
																				comfort			Likart - unspecified		
																				functionality			Likart - unspecified		
																				reliability			Likart - unspecified		
																				usability			Likart - unspecified		
Grimaldi [11]	Human computer design approach ISO 13407:1999 [1] revised by ISO 9241-210:2010	Collaborate		X	X	X	X			X				X		X		X		quality of interaction			Us.E 2.0		
																				satisfaction			Likart - unspecified		
																				usability			Likart - unspecified		
																				errors		Not stated			
Gustafson [14]	NIATx model as a framework Developed recommendations for framework	Empower	X		X	X			X			X			X	X		X	X	usability	Spending time teaching older adults how to use technology in a classroom		Likart - unspecified		

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																			setting					
Harrington [9]	Experience-based co-design Developed recommendations for framework	Empower	X	X	X				X				X					X		usability			Likart - unspecified		
																			usefulness	Likart - unspecified				
Harte [7]	HCD methodology ISO 9241-210 standards	Empower	X	X	X	X				X	X	X		X		X	X	X		cognitive workload			NASA-TLX		X
																				efficiency			ASQ		X
																				satisfaction			Likart - unspecified		
																				usability			SUS		X
																				errors		Mistakes Counted			
																				Performance		Time to Complete			
Hoffman [39]	IPDAS Ottawa Decision Support	Empower		X	X	X	X		X	X	X			X	X	X	X			quality of decision aid			Ottawa Acceptability Scale		X
																				values clarity			SURE		X
																				errors		Observed errors			
Holden [53]	Age-Related Research-Based Usability Guidelines Principles and Creative Human Factors Approaches	Empower		X	X		X			X				X	X		X			usability			SUS		X
																				errors		Mistakes Counted			
																				need for assistance		Times Request Assistance			
																				Performance		Time to Complete			
Kim [38]	Grounded theory approach	Involve	X	X	X				X			X	X		X										
Mansson [35]	Karagianni's Optimized Honeycomb model	Empower					X			X			X			X		X		findable, accessible, usable, desirable, credible, useful, and valuable	UX Honeycomb Model ALL QUALITATIVE				
Martin-Hammond [36]	Nielsen's Heuristics	Involve			X	X					X	X		X		X	X			usability	qualitative				
Nguyen [42]	Medical Research Council framework Developed recommendations for framework	Empower			X		X			X			X		X	X		X		satisfaction			Likart - unspecified		
																				errors		Observed errors			
																				Navigation	Google Analytics				
Nielsen [44]	HCD International Organization for Standardization (ISO) standards for human-centered design for	Empower							X			X			X	X		X		usability			Likart - unspecified		

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	interactive systems (European Committee for Standardization, 2010) and applied principles and techniques from the user innovation management method (Kanstrup & Bertelsen, 2011).																	user acceptance			Likart - unspecified		
Or [43]	Human Factors Design principles Nielson heuristic principles Systematic usability study guideline	Collaborate			X		X			X						X	X			ease of use			Likart - unspecified		
																				satisfaction			Likart - unspecified		
																				usefulness			Likart - unspecified		
																				errors		Mistakes Counted			
																				Need for assistance		Times Request Assistance			
																				Performance		Time to Complete			
Petersen [45]	IPDAS	Empower																		comprehensibility			Likart - unspecified		
					X		X									X		X		effectiveness			Likart - unspecified		
																				satisfaction			Likart - unspecified		
																				usability			Likart - unspecified		
																				value of the information			Likart - unspecified		
Poirier [28]	IPDAS	Consult																		comprehension /recall			Likart - specified validity		X
																				gaze fixation				eye-tracking	
										X	X							X		graphicacy			Likart - specified validity		X
																				numeracy			Likart - specified validity		X
																				values clarity			Decisional Conflict Scale		X
																				performance		Time to Complete			
Portz [40]	Grounded theory approach Human-Centered Design framework	Involve			X		X																		
Scandurra [37]	Human-Computer Interaction (HCI) research theory	Empower											X					X		share experience with others			Likart - unspecified		
																				usability			Likart - unspecified		
																				user acceptance			Likart - unspecified		
Ummels [41]	Double Diamond Method	Empower			X			X	X	X						X		X		satisfaction			Likart - unspecified		

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	MoSCoW model Ummels Framework Developed recommendations for framework																	usability			Likart - unspecified		
Van De Dijk [54]	User Centered Design Approach	Involve		X					X		X		X					X						
Van Velsen [55]	MoSCoW method	Involve									X	X	X		X				usability	qualitative				
Wikberg-Nilsson [8]	None	Empower			X								X			X		X	cognitive process				eye-tracking	
																			usability	qualitative				

A. Contextual Inquiry
B. Survey
C. Interview
D. Cognitive walkthrough
E. Think aloud
F. Remote evaluation
G. Focus group
H. Task analysis
I. Scenarios
J. Prototype (low fidelity)
K. Working Group Workshop
L. Use Cases
M. User Stories
N. Prototype (medium fidelity)
O. Heuristic evaluation
P. Prototype (high fidelity – functional/Beta version)
Q. Other Process

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Supplementary Appendix D

Co-Design Approach	***Inform***		***Consult***		***Involve***		***Collaborate***		***Empower***	
	n	ref	n	ref	n	ref	n	ref	n	ref
Co-creation (n=2)	**2**	[35,41]	**2**	[35,41]	**2**	[35,41]	**2**	[35,41]	**2**	[35,41]
Co-design (n=5)	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]
Human centered computing (n=1)	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]
Human Centered Design (HCD) (n=3)	**3**	[7,11,40]	**3**	[7,11,40]	**3**	[7,11,40]	**2**	[7,11]	**1**	[7]
Human Factors (n=1)	**1**	[33]	**1**	[33]	**1**	[33]	**0**		**0**	
Interaction Design Process (n=1)	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]
Participatory Design (n=7)	**7**	[8,9,34,37,38,44,55]	**7**	[8,9,34,37,38,44,55]	**7**	[8,9,34,37,38,44,55]	**5**	[8,9,34,37,44]	**5**	[8,9,34,37,44]
Usability Evaluation Approach (n=1)	**1**	[43]	**1**	[43]	**1**	[43]	**1**	[43]	**0**	
User Centered Design/Approach (n=9)	**9**	[12,14,29,36,39,41,52–54]	**9**	[12,14,29,36,39,41, 52–54]	**9**	[12,14,29,36,39,41, 52–54]	**5**	[14,29,39,41,53]	**6**	[14,29,39,41,52,53]
Other (Did not Specify) (n=2)	**2**	[28,45]	**2**	[28,45]	**1**	[45]	**1**	[45]	**1**	[45]

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2-4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	4-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	5-6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	4-6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	5
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	NA
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	4-6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	4-6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	5
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	NA
Study characteristics	17	Cite each included study and present its characteristics.	6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	8
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	6-11
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	NA
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	11-13
	23b	Discuss any limitations of the evidence included in the review.	14
	23c	Discuss any limitations of the review processes used.	14
	23d	Discuss implications of the results for practice, policy, and future research.	14
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	15
Competing interests	26	Declare any competing interests of review authors.	15
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71



PRISMA 2020 Checklist

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	No
Registration	12	Provide the register name and registration number.	Yes

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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